RV College of Engineering®



Industrial Engineering & Management

Bachelor of Engineering (B.E)

Scheme And Syllabus of III & IV Semester
(2025 Scheme)

B.E. Programs : CI, AS, BT, CD, CH, CS, CV, CY, EC, EE, ET, IM, IS, ME.
PG Programs (12+1) M.Tech, MCA, M.Sc. (Engg.)
Ph.D. Programs : All Departments are recognized as Research Centres by VTU, Belagavi

2026

RV College of Engineering®

NAAC GRADE A+ ACCREDITED INSTITUTION

SINCE 1963

SHAPING VISIONARIES, CRAFTING INNOVATORS



An autonomous institution affiliated with VTU, Belagavi, and approved by AICTE and UGC, New Delhi

RV SATELLITE-1 (RVSAT - 1) India's First Microbiological Payload, launched by RVCE students in December 2024

NCC - Best Institution Award of Karnataka and Goa Directorate (2019, 2022, 2023 & 2025)

AUTONOMOUS CAR India's first driverless car has been successfully tested at Bengaluru's RV College of Engineering as a part of Wipro-IISc-RVCE WIRIN (Wipro-IISc Research and Innovation Network) initiative.

₹50 Cr. MoU with TATA Technologies to establish Centre for Invention, Innovation, Incubation & Training.

MoU with Bajaj Auto Foundation on to implement the **Bajaj Engineering Skills Training (BEST) Programme**. The initiative will provide industry-focused training and hands-on learning in Electronics Manufacturing Systems (EMS).

Placements

- **₹67 LPA** Highest Package
- **250+** Recruitment Partners
- **92%** Placement Rate
- **₹11 LPA** Median Salary

Rankings

- **101-150** NIRF Rankings 2025 Engineering Category
- **1501+** Times Higher Education World University Rankings-2026
- **601+** Times Higher Education Asia University Rankings-2025
- **NBA** Eligible UG and PG programs accredited multiple times
- **IIRF 2025** Engineering ranking India National - 01 State - 01 Zone - 01
- **NPTEL** Local Chapter Ranking 2025 National - 21 State – 01 **AA Rating**

Research and Centres

37 Skill Labs

General Skills, Domain Skills, Interdisciplinary Skills

- Bosch Rexroth
- Mercedes Benz
- Morris Garage
- Boston AI System
- Toyota Kirloskar
- Vision Astra EV Academy

Outcome Based Education (OBE) emphasizing on experiential learning, research-based learning and skill-based learning

- **20** Centres of Excellence
- **8** Centres of Competence
- **123** Patents filed
- **98** Patents published
- **51** Patents granted
- **32** Innovative & Cultural activity teams
- **₹30 Cr.** Research and projects consultancy (last 5 years)

16 Research Centres in niche areas

- Autonomous Vehicles
- Nanomaterials and Devices
- Hydrogen and Green Technology
- Extended Reality and Visnet Computing
- Healthcare
- E-mobility
- IoT and Sensors
- Smart Antenna System
- Computational Genomics
- Quantum Computing

Programs Intake and Staff

- **13 UG** Programs;**1420 (2025)** annual sanctioned intake
- **13 PG** Programs including MCA **354 (2025)** annual sanctioned intake
- **Ph.D.** Programs in all departments
- **500+** Faculty Members and Staff

State-of-the-Art infrastructure

- **Innovation Center**
- **Student Spaces**
- **Gymnatorium**
- **Hostels**
- **Labs & Workshops**
- **Food Court**

CET / KEA Code: **E005**
COMED-K Code: **E095**
PGCET Code (MTech): **T857**
PGCET Code (MCA): **C463**

Alumni

- **36,000 +** Alumni across the globe
- **100+** RVCE alumni in civil service

RVCE Reconnect - USA edition, our first International Alumni Meet

NUMBER OF ATTENDEES

- **650** San Jose
- **120** Seattle

Mr. Suresh Katta, RVCE alumnus & Chairman Emeritus of Saama Group, USA and **Mr. Anil Kumble,** RVCE alumnus & former Indian cricketer, were the Guests of Honour at the alumni meets held in San Jose & Seattle, USA.

Collaborations with 140+ MoUs with Industry, Institutions & Research Establishments

International universities

thws Technical University of Applied Sciences Würzburg-Schweinfurt · Fachhochschule Dortmund University of Applied Sciences and Arts · Technische Hochschule Rosenheim · HTWD · The University of Iowa · FIU Florida International University · Saint Peter's University · Kennesaw State University · RWTH Aachen University · The University of Arizona

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Go, change the world®

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Scheme And Syllabus of III & IV Semester
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2026

INDUSTRIAL ENGINEERING & MANAGEMENT

DEPARTMENT VISION

Imparting innovation and value-based education in Industrial Engineering and Management for steering organizations to global standards with an emphasis on sustainable and inclusive development.

DEPARTMENT MISSION

1. To impart scientific knowledge, engineering and managerial skills for driving organizations to global excellence.
2. To promote a culture of training, consultancy, research and entrepreneurship interventions among the students.
3. To institute collaborative academic and research exchange programs with national and globally renowned academia, industries and other organizations.
4. To establish and nurture centers of excellence in the niche areas of Industrial and Systems Engineering.

PROGRAM EDUCATIONAL OBJECTIVES

PEO1	Conceive, design, implement and operate integrated systems, focus on appropriate measures of performance at strategic, tactical and operational levels.
PEO2	Develop competency to adapt to changing roles for achieving organizational excellence.
PEO3	Design and develop sustainable technologies and solutions for betterment of society.
PEO4	Pursue entrepreneurial venture with a focus on creativity and innovation for developing newer products, processes and systems.

PROGRAM SPECIFIC OUTCOMES

PSO1	Design, develop, implement and improve integrated systems that include people, Materials, information, equipment and energy.
PSO2	Apply statistical and simulation tools, optimization and meta heuristics techniques for analysis of various systems leading to better decision making.
PSO3	Demonstrate the engineering relationships between the management tasks of planning, Organization, leadership, control, and the human element in various sectors of economy.

LEAD SOCIETY

Institute of Industrial Engineers (IIE)

ABBREVIATIONS

Sl. No.	Abbreviation	Meaning
1.	VTU	Visvesvaraya Technological University
2.	BS	Basic Sciences
3.	CIE	Continuous Internal Evaluation
4.	SEE	Semester End Examination
5.	PE	Professional Core Elective
6.	IE	Institutional Elective
7.	HSS	Humanities and Social Sciences
8.	PY	Physics
9.	CM	Chemistry
10.	MA	Mathematics
11.	AS	Aerospace Engineering
12.	AI&ML	Artificial Intelligence & Machine Learning
13.	BT	Biotechnology
14.	CH	Chemical Engineering
15.	CS	Computer Science & Engineering
16.	CV	Civil Engineering
17.	EC	Electronics & Communication Engineering
18.	EE	Electrical & Electronics Engineering
19.	EI	Electronics & Instrumentation Engineering
20.	ET	Telecommunication Engineering
21.	IM	Industrial Engineering & Management
22.	IS	Information Science & Engineering
23.	ME	Mechanical Engineering
24.	CD	Computer Science & Engineering (Data Science)
25.	CY	Computer Science & Engineering (Cyber Security)
26.	CI	Computer Science & Engineering (Artificial Intelligence & Machine Learning)

Semester - III			
Sl. No.	**Course Code**	**Course Title**	**Page No.**
1	MA231TB	Statistics, Laplace Transform and Numerical Methods	1
2	CV232TA	Environment and Sustainability	3
3	IM233AI	Work Systems Design	6
4	IM334AI	Manufacturing Processes and Automation	9
5	IM335AI	Digital Measurements and Metrology	12
6	IM236AT	Human Resource Management and Analytics	14
7	HS238LX	AE Course	17
8	MAT139BT	Bridge Course Mathematics	34

Semester - IV			
Sl. No.	**Course Code**	**Course Title**	**Page No.**
1	IM341AT	Probability and Applied Statistics	36
2	ME242TB	Material Science for Engineers	39
3	IM243AI	Principles of Ergonomics	41
4	IM244AI	Operations Research	44
5	IM245AT	Marketing Management	47
6	XX246TAX	Professional Electives	49
7	IM248DL	Applied Design Thinking Lab	58
8	CS149BT	Bridge Course C Programming	60

2025 Scheme template – UG Programs													
B.E in Indusrial Engineering and Management (IEM)													
III SEMESTER													
Sl.	**Course Code**	**Course Title**	**BoS**	**Credit Allocation**				**Category**	**CIE Duration (Hrs)**	**Max Marks CIE**		**SEE Duration (Hrs)**	**Max Marks SEE**	
				L	**T**	**P**	**Total**			**Theory**	**Lab**		**Theory**	**Lab**
1	MA231TB	Statistics, Laplace Transform and Numerical Methods	MA	3	1	0	4	Theory	2	100	***	3	100	***
2	CV232TA	Environment and Sustainability	CV	3	0	0	3	Theory	2	100	***	3	100	***
3	IM233AI	Work Systems Design	IM	3	0	1	4	Theory+Lab	2+2	100	50	3+3	100	50
4	IM334AI	Manufacturing Processes and Automation	IM	3	0	1	4	Theory+Lab	2+2	100	50	3+3	100	50
5	IM335AI	Digital Measurements and Metrology	IM	2	0	1	3	Theory+Lab	1.5+2	50	50	2+3	50	50
6	IM236AT	Human Resource Management and Analytics	IM	3	0	0	3	Theory	2	100	***	3	100	***
7	HS238LX	AE Course	HS	0	0	1	1	Lab	1	***	50	2	***	50
8	MAT139BT	Bridge Course Mathematics	MA	2	0	0	0*	Theory	1.5	50	***	—	***	***
			Total	**19**	**1**	**4**	**22**							

Note: 1 Credit = 30 hrs (L:T:P)

* Audit

2025 Scheme template – UG Programs													
B.E in Indusrial Engineering and Management (IEM)													
IV SEMESTER													
Sl.	**Course Code**	**Course Title**	**BoS**	**Credit Allocation**				**Category**	**CIE Duration (Hrs)**	**Max Marks CIE**		**SEE Duration (Hrs)**	**Max Marks SEE**	
				L	**T**	**P**	**Total**			**Theory**	**Lab**		**Theory**	**Lab**
1	IM341AT	Probability and Applied Statistics	IM	3	0	0	3	Theory	2	100	***	3	100	***
2	ME242TB	Material Science for Engineers	ME	3	0	0	3	Theory	2	100	***	3	100	***
3	IM243AI	Principles of Ergonomics	IM	3	0	1	4	Theory+Lab	2+2	100	50	3+3	100	50
4	IM244AI	Operations Research	IM	3	0	1	4	Theory+Lab	2+2	100	50	3+3	100	50
5	IM245AT	Marketing Management	IM	3	0	0	3	Theory	2	100	***	3	100	***
6	XX246TAX	Professional Electives	XX	3	0	0	3	Theory	2	100	***	3	100	***
7	IM248DL	Applied Design Thinking Lab	IM	0	0	1	1	Lab	1.5	***	50	2	***	50
8	CS149BT	Bridge Course C Programming	CS	2	0	0	0*	Theory	1.5	50	***	—	***	***
			Total	**20**	**0**	**3**	**21**							

Note: 1 Credit = 30 hrs (L:T:P)

* Audit

Semester - III			
AE Courses			
Sl. No.	**Course Code**	**Course Title**	**Page No.**
1	HS238LA	National Service Scheme	17
2	HS238LB	National Cadet Corps	19
3	HS238LC	Physical Education: Sports and Athletics	21
4	HS238LD	Music	23
5	HS238LE	Dance	25
6	HS238LF	Theatre (Light Camera and Action)	28
7	HS238LG	Art Work and Painting	30
8	HS238LH	Photography and Film Making	32

Semester - IV			
Professional Electives			
GROUP-A			
Sl. No.	**Course Code**	**Course Title**	**Page No.**
1	ME246TAA	Semiconductor Manufacturing and Process Technology	49
2	IM246TAB	Facilities Planning and Material Handling Systems	51
3	IM246TAC	Fundamentals of Sustainability Engineering and Circular Economy	54
4	IM246TAD	Database Management Systems	56

Semester - III

Statistics, Laplace Transform and Numerical Methods
Category: Professional Core Course
(Common to AS, BT, CH, IM, ME)
(Theory)

Course Code	**:**	MA231TB	**CIE**	**:**	100 Marks
Credit distribution L:T:P	**:**	3:1:0	**SEE**	**:**	100 Marks
Pedagogical Hours (L:T:P:S)	**:**	45:30:0:45	**CIE Duration**	**:**	2 Hours
Total credits:	**:**	4	**SEE Duration**	**:**	3 Hours

Unit - I **9 Hours**

Statistics: Central moments, mean, variance, coefficients of skewness and kurtosis in terms of moments. Correlation analysis, rank correlation, curve fitting, linear and multivariate regression analysis.

Unit - II **9 Hours**

Complex Analysis: Complex function, analytic function, Cauchy-Riemann equations, harmonic functions. Construction of analytic function– Milne -Thomson method. Taylor, Laurent series. Zeros and poles, Residue theorem.

Unit - III **9 Hours**

Laplace Transform: Existence and uniqueness of Laplace transform, transform of elementary functions, region of convergence. Properties - linearity, scaling, s - domain shift, differentiation in the s - domain, division by t, differentiation and integration in the time domain. Laplace transform of time domain periodic functions, Heaviside unit step function, unit impulse function, t - shift property.

Unit - IV **9 Hours**

Inverse Laplace Transform: Definition, properties, evaluation using different methods. Convolution theorem. Application to solve ordinary linear differential equations.

Unit - V **9 Hours**

Numerical Methods for Partial Differential Equations: Numerical solutions to partial differential equations – Finite difference approximation to derivatives, solution of Laplace equation in two-dimension, heat and wave equations in one dimension (explicit methods).

Course Outcomes: After completing the course, the students will be able to:

CO1: Explain the fundamental concepts of statistical measures, complex analysis, Laplace transforms, inverse Laplace transforms, and numerical methods for partial differential equations used in scientific and engineering problems.

CO2: Apply appropriate mathematical techniques involving statistics, complex variables, Laplace transforms, inverse transforms, and finite difference methods to formulate and solve engineering problems.

CO3: Analyze and validate engineering solutions using residue calculus, convolution theorem, transform-domain methods, regression analysis, and numerical techniques for ordinary and partial differential equations.

CO4: Integrate and evaluate concepts of statistics, complex analysis, Laplace transform techniques, and numerical methods to model engineering systems, interpret results, and justify mathematical approaches for real-world applications.

References

[1] J. W. Brown and R. V. Churchill, *Complex Variables and Applications*, 8th ed. McGraw-Hill Higher Education, 2004, ISBN: 9780703051942

[2] D. G. Zill and W. S. Wright, *Advanced Engineering Mathematics*, 7th ed. Jones and Bartlett Learning, 2020, ISBN: 9781284105902

[3] J. L. Schiff, *The Laplace Transform: Theory and Applications*. Springer-Verlag New York, 2013, ISBN: 9781475772623

[4] B. S. Grewal, *Higher Engineering Mathematics*, 44th ed. Khanna Publishers, 2015, ISBN: 8174091955

RUBRICS FOR THE CONTINUOUS INTERNAL EVALUATION		
#	**COMPONENTS**	**MARKS**
1	**QUIZZES:** Quizzes will be conducted in online/offline mode. **TWO QUIZZES** will be conducted and each Quiz will be evaluated for 10 Marks. **THE SUM OF TWO WILL BE THE FINAL QUIZ MARKS.**	**20**
2	**TESTS:** Students will be evaluated in test, descriptive questions with different complexity levels (Based on Revised Bloom's Taxonomy Levels).**THREE** tests will be con	**40**
3	**EXPERIENTIAL LEARNING:** Students will be evaluated for their creativity and practical implementation of the problem. Case stufy-based teaching learning (05), Program specific requirements (05), Video based seminar/presentation/demonstration (10), Tools usage (20). **ADDING UPTO 40 MARKS.**	**40**
	MAXIMUM MARKS FOR THE CIE	**100**

RUBRICS FOR SEMESTER END EXAMINATION (THEORY)		
Q.NO.	**CONTENTS**	**MARKS**
	PART A	
1	Objective type questions covering entire syllabus	**10**
	PART B (Maximum of THREE Sub-divisions only)	
2	Unit - I: Compulsory	**18**
3	Unit - II: Compulsory	**18**
4 & 5	Unit - III: Question 4 or 5	**18**
6 & 7	Unit - IV: Question 6 or 7	**18**
8 & 9	Unit - V: Question 8 or 9	**18**
	TOTAL	**100**

Semester - III					
Environment and Sustainability **Category: Professional Core Course** **(Common to all Programs)** **(Theory)**					
Course Code	**:**	CV232TA	**CIE**	**:**	100 Marks
Credit distribution L:T:P	**:**	3:0:0	**SEE**	**:**	100 Marks
Pedagogical Hours (L:T:P:S)	**:**	45:0:0:45	**CIE Duration**	**:**	2 Hours
Total credits:	**:**	3	**SEE Duration**	**:**	3 Hours
Unit - I					**9 Hours**
Environment and Ecosystem: Definition, scope and importance of environment; components of environment; need for environmental awareness. Ecosystem structure and functions, energy flow, ecological succession, biodiversity and ecosystem services. Natural resources, environmental degradation, climate change and global environmental issues.					
Unit - II					**9 Hours**
Natural Resources and Environmental Management: Water demand and conservation, rainwater harvesting and runoff estimation. Municipal solid waste, hazardous waste and e-waste management. Waste minimization techniques, landfill concepts and numerical problems on waste generation. Sustainable utilization of water, land and energy resources.					
Unit - III					**9 Hours**
Sustainable Development and Circular Economy: Concept of development and sustainability; economic, social and environmental dimensions; challenges to sustainability. Millennium Development Goals (MDGs) and Sustainable Development Goals (SDGs). Systems thinking, circular economy, industrial ecology and life-cycle thinking. Applications to water, energy, food, land, forests and waste management.					
Unit - IV					**9 Hours**
Green Technologies and Emerging Tools for Sustainability: Energy management and conservation. Hydrogen, ocean, tidal and geothermal energy. Sustainable urbanization, green buildings and green materials. AI for climate modelling, biodiversity assessment, pollution monitoring using sensors and satellite data, AI & GIS for resource mapping, smart water management, air quality forecasting, disaster prediction and smart waste management.					
Unit - V					**9 Hours**
Environmental Governance, Safety and CSR: Environmental protection and environmental legislation. Environmental audit. Occupational Health and Safety Management Systems (OHSMS), ISO 45001:2018. Fire safety, prevention and firefighting systems. Corporate Social Responsibility (CSR), stakeholder management, sustainability reporting (GRI, DJSI, CEPI) and investor perspectives.					
Course Outcomes: After completing the course, the students will be able to:					
CO1: Explain the concepts of environment, ecosystems, natural resources and environmental management practices for sustainable development.					
CO2: Analyze sustainability principles, circular economy concepts, SDGs and resource management strategies for addressing environmental challenges.					
CO3: Apply renewable energy technologies, green engineering practices and AI-enabled tools for environmental monitoring and sustainable resource utilization.					
CO4: Interpret environmental legislation, occupational health and safety requirements, corporate social responsibility and sustainability reporting frameworks for responsible environmental governance.					
References					

[1] B. Joseph, *Environmental Science and Engineering*, 3rd ed. New Delhi: Tata McGraw-Hill Education, 2017, ISBN: 9789387432352

[2] G. M. Masters and W. P. Ela, *Introduction to Environmental Engineering and Science*, 3rd ed. Upper Saddle River, NJ: Pearson Education, 2008, ISBN: 9780132339346

[3] H. S. Peavy et al., *Environmental Engineering*. New York: McGraw-Hill, 1985, ISBN: 9789351340263

[4] D. Crowther and G. Aras, Eds., *A Handbook of Corporate Governance and Social Responsibility*. Aldershot, UK: Gower Publishing, 2008, ISBN: 9780566088179

[5] P. Hawken, *The Ecology of Commerce: A Declaration of Sustainability*, Revised. New York: HarperBusiness, 2010, ISBN: 9780061252792

[6] Ministry of Environment, Forest and Climate Change, *Environmental impact assessment notification, 2006*, Including amendments, Office Memoranda (OMs), and Gazette Notifications, 2006

[7] United Nations, *Transforming our world: The 2030 agenda for sustainable development*, 2015

[8] International Organization for Standardization, *Iso 45001:2018: Occupational health and safety management systems – requirements with guidance for use*, 2018

RUBRICS FOR THE CONTINUOUS INTERNAL EVALUATION		
#	**COMPONENTS**	**MARKS**
1	**QUIZZES:** Quizzes will be conducted in online/offline mode. TWO quizzes will be conducted and each Quiz will be evaluated for 10 Marks. **The sum of the two will be the final quiz marks.**	**20**
2	**TESTS:** Students will be evaluated in test through descriptive questions with different complexity levels (Revised Bloom's Taxonomy Levels: Remembering, Understanding, Applying, Analyzing, Evaluating, and Creating). THREE tests will be conducted (Two regular tests and One optional Improvement test). Each test will be evaluated for 50 Marks, adding upto 100 Marks. **Final test marks will be reduced to 40**.	**40**
3	**EXPERIENTIAL LEARNING:** Students shall be evaluated based on the creativity, innovation, and practical implementation of their proposed solutions. The evaluation shall be conducted in two phases. The implementation modalities for experiential learning activity shall be decided considering the nature of the course and its interdisciplinary requirements. **Each phase shall carry 20 marks, with the total evaluation carrying 40 marks.**	**40**
	MAXIMUM MARKS FOR THE CIE	**100**

RUBRICS FOR SEMESTER END EXAMINATION (THEORY)		
Q.NO.	**CONTENTS**	**MARKS**
	PART A	
1	Objective type questions covering entire syllabus	**10**
	PART B (Maximum of THREE Sub-divisions only)	
2	Unit - I: Compulsory	**18**
3	Unit - II: Compulsory	**18**

RUBRICS FOR SEMESTER END EXAMINATION (THEORY) (Continued)		
Q.NO.	**CONTENTS**	**MARKS**
4 & 5	Unit - III: Question 4 or 5	**18**
6 & 7	Unit - IV: Question 6 or 7	**18**
8 & 9	Unit - V: Question 8 or 9	**18**
	TOTAL	**100**

Semester - III					
Work Systems Design **Category: Professional Core Course** **(Theory & Practical)**					
Course Code	**:**	IM233AI	**CIE**	**:**	100+ 50 Marks
Credit distribution L:T:P	**:**	3:0:1	**SEE**	**:**	100+ 50 Marks
Pedagogical Hours (L:T:P:S)	**:**	45:0:30:45	**CIE Duration**	**:**	2+2 Hours
Total credits:	**:**	4	**SEE Duration**	**:**	3+3 Hours

Unit - I	**9 Hours**
Introduction to Work Systems and Productivity: Pyramidal structure of work, Importance of Time, Physical work systems, work system as a field of professional practice, productivity (includes numericals). **Work and Worker-Machine Systems:** Manual work systems, Worker-machine systems and Automated work systems. Numericals on cycle time analysis of manual work and worker-machine systems.	
Unit - II	**9 Hours**
Introduction to Methods Engineering and Operations Analysis: Evolution and scope of methods engineering, Application of Methods engineering, Basic Data collection and Analysis techniques (includes case problems), Methods engineering and automation. **Charting and Diagramming techniques for Operations Analysis:** Overview, Network diagrams, Traditional IE charting and Diagramming techniques, Block diagrams and process maps. principles of motion economy and work design. AI in Methods Engineering & Productivity.	
Unit - III	**9 Hours**
Introduction to Work Measurement: Determination of Time Standards, Prerequisites for valid time standards, Allowances in Time standards, Methods of determining time standards ranked by relative accuracy, Methods of determining time standards ranked by relative application speed. **Direct Time Study:** Procedure for standard time determination, Numericals on determining time standards for pure manual tasks and worker machine tasks. Predetermined Motion Time Systems: Conceptual overview of PMTS and MTM, Basic MOST, Additional versions of MOST, MOST for windows. Intelligent Work Measurement.	
Unit - IV	**8 Hours**
Lean Production - Elimination of waste: Production of defective parts, over production and excessive inventories, other forms of waste. Just in Time Production: Pull system of production control, set up time reduction, stable and reliable production operations. **Autonomation:** Stopping the process, error prevention and Total productive maintenance. **Worker Involvement:** Continuous improvement, Visual management and 5S, Standardized work procedures (Conceptual Treatment only). AI driven Lean Systems.	
Unit - V	**8 Hours**
Introduction to Ergonomics and Human Factors: Overview of Ergonomics, Understanding the interactions in Human –Machine system, Topic Areas in Ergonomics-Physical Ergonomics, Cognitive Ergonomics, Physical Work Environment. **Occupational Safety and Health:** Industrial Accidents and Injuries, Occupational Disorders and diseases. AI Wearables for Safety, Human-AI Collaboration.	

Laboratory Component

1. Exercises based on Methods Engineering
2. Exercises based on Work Measurement
3. Exercises based on Physical, Cognitive and Environmental Ergonomics

Course Outcomes: After completing the course, the students will be able to:

CO1: Recognize the relevance of industrial engineering principles for productivity improvement in the organizations.

CO2: Design and optimize lean work processes by integrating traditional charting techniques with digital process mapping.

CO3: Establish time standards for manual and machine tasks using engineering work measurement principles.

CO4: Summarize the importance of human-centric workstations to enhance worker safety and organizational performance.

References

[1] M. P. Groover, *Work Systems: The Methods, Measurement, and Management of Work*. Pearson India Education, 2017, ISBN: 9789332581241

[2] G. Kanawaty, *Introduction to Work Study*, 4th ed. Geneva: International Labour Office (ILO), 1992, ISBN: 9221071081

[3] F. E. Meyers and J. R. Stewart, *Motion and Time Study for Lean Manufacturing*, 3rd ed. Prentice Hall, 2002, ISBN: 0130316709

[4] M. S. Sanders and E. J. McCormick, *Human Factors in Engineering and Design*, 7th ed. McGraw-Hill, 1993, ISBN: 0070449023

[5] G. Salvendy, Ed., *Handbook of Industrial Engineering: Technology and Operations Management*, 3rd ed. John Wiley & Sons, 2001, ISBN: 9780471330577

[6] *Handbook of industrial engineering, 3rd edition (online copy)*, `https://pdfcoffee.com/handbook-of-industrial-engineering-3rd-editionpdf-pdf-free.html`, Accessed online

RUBRICS FOR THE CONTINUOUS INTERNAL EVALUATION

#	COMPONENTS	MARKS
1	**QUIZZES:** Quizzes will be conducted in online/offline mode. TWO quizzes will be conducted and each Quiz will be evaluated for 10 Marks. **The sum of the two will be the final quiz marks.**	**20**
2	**TESTS:** Students will be evaluated in test through descriptive questions with different complexity levels (Revised Bloom's Taxonomy Levels: Remembering, Understanding, Applying, Analyzing, Evaluating, and Creating). THREE tests will be conducted (Two regular tests and One optional Improvement test). Each test will be evaluated for 50 Marks, adding upto 100 Marks. **Final test marks will be reduced to 40**.	**40**
3	**EXPERIENTIAL LEARNING:** Students shall be evaluated based on the creativity, innovation, and practical implementation of their proposed solutions. The evaluation shall be conducted in two phases. The implementation modalities for experiential learning activity shall be decided considering the nature of the course and its interdisciplinary requirements. **Each phase shall carry 20 marks, with the total evaluation carrying 40 marks.**	**40**
4	**LAB:** Conduction of laboratory exercises, lab report, observation and analysis (30 Marks), Lab test (10 Marks) and problem based learning review phase I and II (10 Marks) adding upto 50 marks . **The final lab marks is for 50.**	**50**
	MAXIMUM MARKS FOR THE CIE	**150**

RUBRICS FOR SEMESTER END EXAMINATION (THEORY)		
Q.NO.	**CONTENTS**	**MARKS**
	PART A	
1	Objective type questions covering entire syllabus	**10**
	PART B (Maximum of THREE Sub-divisions only)	
2	Unit - I: Compulsory	**18**
3	Unit - II: Compulsory	**18**
4 & 5	Unit - III: Question 4 or 5	**18**
6 & 7	Unit - IV: Question 6 or 7	**18**
8 & 9	Unit - V: Question 8 or 9	**18**
	TOTAL	**100**

RUBRICS FOR SEMESTER END EXAMINATION (LAB)		
Q.NO.	**CONTENTS**	**MARKS**
1	Evaluation will be conducted as problem / project based learning, which is program / laboratory specific	**50**
	TOTAL	**50**

Semester - III					
Manufacturing Processes and Automation **Category: Professional Core Course** **(Theory & Practical)**					
Course Code	**:**	IM334AI	**CIE**	**:**	100+ 50 Marks
Credit distribution L:T:P	**:**	3:0:1	**SEE**	**:**	100+ 50 Marks
Pedagogical Hours (L:T:P:S)	**:**	45:0:30:45	**CIE Duration**	**:**	2+2 Hours
Total credits:	**:**	4	**SEE Duration**	**:**	3+3 Hours

Unit - I	**7 Hours**
Manufacturing Processes: Recent Developments in Manufacturing, Metal Casting Process: Casting terminology, sand mould making procedure, Pattern allowance, Pattern materials, and types of patterns, Molding sand properties. **Special Casting Processes:** Principle of extrusion. Metal spinning and hydro forming techniques, Shell molding, Rolling, forging operations	
Unit - II	**7 Hours**
Theory Of Metal Cutting: Single point tool nomenclature, geometry, orthogonal & oblique cutting, Mechanism of chip formation, types of chips, Tool wear, Tool life criteria, Taylor's tool life equation, problems on tool life evaluation, Machinability, factors affecting machinability	
Unit - III	**9 Hours**
Numerical & Computer control in Production system: NC procedure, NC coordinate systems, Elements & Classification of NC system, Functions & Features of CNC, DNC Concepts, and Components & Types of DNC. **NC part programming & computer aided part programming:** Manual part programming, Computer Assisted part programming, Computer assisted NC part programming.	
Unit - IV	**7 Hours**
Automation: Introduction, Definition of Automation, Mechanization vs. Automation, Advantages of Automation, Goals of Automation, Social Issues of Automation, Low-Cost Automation, Types of Automation Current Emphasis in Automation, Reasons for Automation, Reasons for not Automation, Strategies for Automation.	
Unit - V	**8 Hours**
Robotics: Introduction, History of Robots, Industrial Robot, Laws of Robotics Motivating Factors, Characteristics of an Industrial Robot, Components of an Industrial Robot, Robot Wrist and End of Arm Tools, Robot Terminology, Robotic Joints. **Robot End Effectors & Robot program:** Introduction to End Effector, Classification of End Effectors, Grippers, Selection of Gripper, Gripping Mechanisms. Introduction to Robot Programming, Robot Programming Techniques	

Part – I

1. Preparation of models involving the following lathe operations: Plain Turning, Taper Turning, Step Turning, Thread Cutting, Facing, Knurling and forming. Cutting of gear teeth using milling machine

Part – II

2. Two experiments on Simulation of Turning and milling operation on CNC Train software. Four experiments on CNC turning & milling machines.

Course Outcomes: After completing the course, the students will be able to:

CO1: Explain the principles of manufacturing processes, metal cutting, numerical control systems, automation and industrial robotics used in modern production environments.

CO2: Analyze manufacturing, machining and automated production systems to determine their performance, productivity and suitability for Industrial applications.

CO3: Apply manufacturing technologies, CNC programming concepts, automation strategies and robotic programming techniques to solve production and manufacturing problems.

CO4: Evaluate and recommend appropriate manufacturing, CNC, automation and robotic systems for achieving quality, productivity and cost-effective production objectives.

References

[1] J. P. Kaushish, *Manufacturing Processes*, 2nd ed. PHI Learning Pvt. Ltd., 2010, ISBN: 9788120340824

[2] M. P. Groover, *Automation, Production Systems, and Computer-Integrated Manufacturing*, 3rd ed. New Delhi: Prentice Hall of India (PHI), 2007, ISBN: 9780132393218

[3] K. L. Narayan et al., *Computer Aided Design and Manufacturing*, 1st ed. New Delhi: Prentice Hall of India (PHI), 2008, ISBN: 9788120333420

[4] A. K. Gupta and S. K. Arora, *Industrial Automation and Robotics*, 3rd ed. University Science Press, 2013

RUBRICS FOR THE CONTINUOUS INTERNAL EVALUATION		
#	**COMPONENTS**	**MARKS**
1	**QUIZZES:** Quizzes will be conducted in online/offline mode. TWO quizzes will be conducted and each Quiz will be evaluated for 10 Marks. **The sum of the two will be the final quiz marks.**	**20**
2	**TESTS:** Students will be evaluated in test through descriptive questions with different complexity levels (Revised Bloom's Taxonomy Levels: Remembering, Understanding, Applying, Analyzing, Evaluating, and Creating). THREE tests will be conducted (Two regular tests and One optional Improvement test). Each test will be evaluated for 50 Marks, adding upto 100 Marks. **Final test marks will be reduced to 40**.	**40**
3	**EXPERIENTIAL LEARNING:** Students shall be evaluated based on the creativity, innovation, and practical implementation of their proposed solutions. The evaluation shall be conducted in two phases. The implementation modalities for experiential learning activity shall be decided considering the nature of the course and its interdisciplinary requirements. **Each phase shall carry 20 marks, with the total evaluation carrying 40 marks.**	**40**
4	**LAB:** Conduction of laboratory exercises, lab report, observation and analysis (30 Marks), Lab test (10 Marks) and problem based learning review phase I and II (10 Marks) adding upto 50 marks . **The final lab marks is for 50.**	**50**
	MAXIMUM MARKS FOR THE CIE	**150**

RUBRICS FOR SEMESTER END EXAMINATION (THEORY)		
Q.NO.	**CONTENTS**	**MARKS**
	PART A	
1	Objective type questions covering entire syllabus	**10**
	PART B (Maximum of THREE Sub-divisions only)	
2	Unit - I: Compulsory	**18**
3	Unit - II: Compulsory	**18**

RUBRICS FOR SEMESTER END EXAMINATION (THEORY) (Continued)		
Q.NO.	**CONTENTS**	**MARKS**
4 & 5	Unit - III: Question 4 or 5	**18**
6 & 7	Unit - IV: Question 6 or 7	**18**
8 & 9	Unit - V: Question 8 or 9	**18**
	TOTAL	**100**

RUBRICS FOR SEMESTER END EXAMINATION (LAB)		
Q.NO.	**CONTENTS**	**MARKS**
1	Evaluation will be conducted as problem / project based learning, which is program / laboratory specific	**50**
	TOTAL	**50**

Semester - III					
Digital Measurements and Metrology **Category: Professional Core Course** **(Theory & Practical)**					
Course Code	**:**	IM335AI	**CIE**	**:**	50+ 50 Marks
Credit distribution L:T:P	**:**	2:0:1	**SEE**	**:**	50+ 50 Marks
Pedagogical Hours (L:T:P:S)	**:**	30:0:30:30	**CIE Duration**	**:**	1.5+2 Hours
Total credits:	**:**	3	**SEE Duration**	**:**	2+3 Hours

Unit - I	**10 Hours**

Measurement Systems, Sensors and Transducers: Concept of measurements and generalized measurement system. Static and dynamic characteristics of measuring instruments, Errors in measurement and statistical evaluation of measurement data, Standards of measurement and calibration, Sensors and transducers: definitions and specifications Classification and selection of transducers Resistive, capacitive and inductive transducers Piezoelectric, Hall effect, optical and digital transducers.

Unit - II	**10 Hours**

Sensors, Data Acquisition and Digital Measurement Systems: Classification of sensors: Displacement, position and proximity sensors (potentiometers), Velocity and motion sensors (tachogenerators), Force and pressure sensors (strain gauge load cell, piezoelectric sensors), Pyroelectric and tactile sensors. Elements of Data Acquisition System (DAS), Analog to Digital (A/D) and Digital to Analog (D/A) converters, Digital measurement systems and computer-aided inspection.

Unit - III	**10 Hours**

Engineering Metrology, GD&T and Advanced Measurement Techniques: Limits, fits and tolerances: Definition of tolerance and limits of size, Hole basis and shaft basis systems, Types of fits and their designation (IS 919–1963), Principle of interchangeability and selective assembly, Introduction to Geometric Dimensioning and Tolerancing (GD&T),Optical and form measurements: Interferometry, optical flats and tool maker's microscope, Measurement of screw threads and gears, Measurement of surface finish and form errors. Laser metrology and interferometric measurements, Geometrical checks on machine tools, Coordinate Measuring Machine (CMM): construction, types and applications.

Laboratory Component

1. Exercises based on Measurement
2. Exercises based on Calibration

Course Outcomes: After completing the course, the students will be able to:

CO1: Discuss the principles and practices of metrology in manufacturing environment and analyze uncertainty in an appropriate manner

CO2: Describe the operating principles of range of widely used instrumentation techniques and illustrate how to use them in the design of measurement systems.

CO3: Compare the production process, the product function and the product design, and to select appropriate measurement quantities and tools for these purposes.

CO4: Evaluate and respond to the need for rigorous and formal metrology concepts in designing and using measurement systems

References

[1] R. K. Jain, *Engineering Metrology*, 18th ed. Khanna Publishers, 2006, ISBN: 817409024X

[2] T. G. Beckwith and N. L. Buck, *Mechanical Measurements*, 5th ed. Addison-Wesley, 1991, ISBN: 8178080559

[3] A. K. Sawhney, *Electrical and Electronic Measurements and Instrumentation*, 18th ed. Dhanpat Rai and Sons, 2008, ISBN: 8177000160

[4] S. Beeby, *MEMS Mechanical Sensors*. Artech House, 2004, ISBN: 1580535364

RUBRICS FOR THE CONTINUOUS INTERNAL EVALUATION		
#	**COMPONENTS**	**MARKS**
1	**QUIZZES:** Quizzes will be conducted in online/offline mode. TWO quizzes will be conducted and each Quiz will be evaluated for 10 Marks. **The average of two will be the final quiz marks.**	**10**
2	**TESTS:** Students will be evaluated in test through descriptive questions with different complexity levels (Revised Bloom's Taxonomy Levels: Remembering, Understanding, Applying, Analyzing, Evaluating, and Creating). THREE tests will be conducted (Two regular tests and One optional Improvement test). Each test will be evaluated for 25 Marks, adding upto 50 Marks. **Final test marks will be reduced to 20**.	**20**
3	**EXPERIENTIAL LEARNING:** Students shall be evaluated based on the creativity, innovation, and practical implementation of their proposed solutions. The evaluation shall be conducted in two phases. The implementation modalities for experiential learning activity shall be decided considering the nature of the course and its interdisciplinary requirements. **Each phase shall carry 10 marks, with the total evaluation carrying 20 marks.**	**20**
4	**LAB:** Conduction of laboratory exercises, lab report, observation and analysis (30 Marks), Lab test (10 Marks) and problem based learning review phase I and II (10 Marks) adding upto 50 marks . **The final lab marks is for 50.**	**50**
	MAXIMUM MARKS FOR THE CIE	**100**

RUBRICS FOR SEMESTER END EXAMINATION (THEORY)		
Q.NO.	**CONTENTS**	**MARKS**
	PART A	
1	Objective type questions covering entire syllabus	**10**
	PART B (Maximum of THREE Sub-divisions only)	
2	Unit - I: Compulsory	**12**
3 & 4	Unit - II: Question 3 or 4	**14**
5 & 6	Unit - III: Question 5 or 6	**14**
	TOTAL	**50**

RUBRICS FOR SEMESTER END EXAMINATION (LAB)		
Q.NO.	**CONTENTS**	**MARKS**
1	Evaluation will be conducted as problem / project based learning, which is program / laboratory specific	**50**
	TOTAL	**50**

Semester - III					
Human Resource Management and Analytics **Category: Professional Core Course** **(Theory)**					
Course Code	**:**	IM236AT	**CIE**	**:**	100 Marks
Credit distribution L:T:P	**:**	3:0:0	**SEE**	**:**	100 Marks
Pedagogical Hours (L:T:P:S)	**:**	45:0:0:45	**CIE Duration**	**:**	2 Hours
Total credits:	**:**	3	**SEE Duration**	**:**	3 Hours

Unit - I	**9 Hours**
Introduction to Management and Organization: Managers and Management (Basic Concepts), Levels of Management, Managerial roles, Functions of Management, Organizations and their Characteristics, Management in Today's Organizations, Functional Areas of Management.**(Not for Examination) Introduction to Human Resource Management:** Meaning, Nature, Scope and Objectives of HRM, Importance of HRM in Organizations, Functions of HRM, Line and Staff Aspects of HRM, Role and Responsibilities of HR Manager, Competencies of HR Professionals. **Job Analysis and Job Design:** Concept and Importance of Job Analysis, Job Analysis Process, Methods of Collecting Job Analysis Information, Job Description and Job Specification, Introduction to Job Design.	
Unit - II	**9 Hours**
Human Resource Planning and Talent Acquisition Human Resource Planning: Meaning and Objectives of Human Resource Planning, Manpower Planning Process, Demand and Supply Forecasting (Conceptual), Succession Planning. **Recruitment:** Meaning and Importance of Recruitment, Recruitment Process, Internal and External Sources of Recruitment, Factors Affecting Recruitment. **Selection and Interviewing:** Selection Process, Types of Selection Tests, Reliability and Validity of Tests (Basic Concept), Interviews – Types and Process, Placement and Induction.	
Unit - III	**9 Hours**
Training, Development and Performance Appraisal Training and Development: Meaning and Importance of Training, Training Process, Methods of Training, Management Development Programs, Training Needs Identification. **Training Evaluation:** Evaluation of Training Effectiveness, Models of Training Evaluation (Conceptual), Benefits of Training to Organization and Employees. **Performance Appraisal:** Concept and Objectives of Performance Appraisal, Traditional and Modern Methods of Performance Appraisal, Appraisal Interview, Errors in Performance Appraisal.	
Unit - IV	**9 Hours**
Introduction to HR Analytics: Meaning and Need for HR Analytics, Evolution of HR Analytics, Role of HR Analytics in Decision Making, Difference between Traditional HR and Data-driven HR. **HR Data and Metrics:** Types of HR Data, Sources of HR Data, Introduction to HR Metrics, Importance of HR Metrics in HR Decision Making. **Types of HR Analytics:** Descriptive Analytics, Diagnostic Analytics – Meaning and Applications in HR (Conceptual Understanding).	
Unit - V	**9 Hours**
HR Metrics, Reporting and Basic Analytics Applications HR Metrics and Measurement: Key HR Metrics – Attrition Rate, Absenteeism Rate, Turnover Rate, Training Cost, Productivity Measures (Basic Formulas and Interpretation). **HR Reporting and Visualization:** HR Reports, HR Scorecards, Basic HR Dashboards (Conceptual), Use of Charts and Graphs for HR Data Presentation. **Ethical Issues in HR Analytics:** Data Privacy, Confidentiality of Employee Data, Ethical Use of HR Data, Introduction to Bias in HR Decision Making.	

Course Outcomes: After completing the course, the students will be able to:

CO1: Understand the fundamental concepts, functions, and roles of Human Resource Management in organizations.

CO2: Explain the processes of human resource planning, recruitment, selection, and induction.

CO3: Describe training, development, and performance appraisal systems used in organizations.
CO4: Understand basic HR analytics concepts, HR metrics, and interpret simple HR data for decision-making.

References

[1] S. P. Robbins et al., *Management*, 15th ed. Pearson Education, 2021, Recommended for Unit-0, ISBN: 9780135581858

[2] G. Dessler and B. Varkkey, *Human Resource Management*, 17th ed. Pearson Education, 2023, Indian Edition; Recommended for Units I–III, ISBN: 9789357054287

[3] K. Aswathappa, *Human Resource Management: Text and Cases*, 10th ed. McGraw Hill Education, 2023, Recommended for Units I–III, ISBN: 9789355323675

[4] D. K. Bhattacharyya, *HR Analytics: Understanding Theories and Applications*, 2nd ed. SAGE Publications, 2023, Recommended for Units IV–V

[5] J. P. Isson and J. S. Harriott, *People Analytics in the Era of Big Data: Changing the Way You Attract, Acquire, Develop, and Retain Talent*. John Wiley & Sons, 2016, Recommended for Units IV–V, ISBN: 9781119050780

[6] M. R. Edwards and K. Edwards, *Predictive HR Analytics: Mastering the HR Metric*, 2nd ed. Kogan Page, 2019, Recommended for Units IV–V, ISBN: 9780749484449

RUBRICS FOR THE CONTINUOUS INTERNAL EVALUATION		
#	**COMPONENTS**	**MARKS**
1	**QUIZZES:** Quizzes will be conducted in online/offline mode. TWO quizzes will be conducted and each Quiz will be evaluated for 10 Marks. **The sum of the two will be the final quiz marks.**	**20**
2	**TESTS:** Students will be evaluated in test through descriptive questions with different complexity levels (Revised Bloom's Taxonomy Levels: Remembering, Understanding, Applying, Analyzing, Evaluating, and Creating). THREE tests will be conducted (Two regular tests and One optional Improvement test). Each test will be evaluated for 50 Marks, adding upto 100 Marks. **Final test marks will be reduced to 40**.	**40**
3	**EXPERIENTIAL LEARNING:** Students shall be evaluated based on the creativity, innovation, and practical implementation of their proposed solutions. The evaluation shall be conducted in two phases. The implementation modalities for experiential learning activity shall be decided considering the nature of the course and its interdisciplinary requirements. **Each phase shall carry 20 marks, with the total evaluation carrying 40 marks.**	**40**
	MAXIMUM MARKS FOR THE CIE	**100**

RUBRICS FOR SEMESTER END EXAMINATION (THEORY)		
Q.NO.	**CONTENTS**	**MARKS**
	PART A	
1	Objective type questions covering entire syllabus	**10**
	PART B (Maximum of THREE Sub-divisions only)	
2	Unit - I: Compulsory	**18**

RUBRICS FOR SEMESTER END EXAMINATION (THEORY) (Continued)		
Q.NO.	**CONTENTS**	**MARKS**
3	Unit - II: Compulsory	**18**
4 & 5	Unit - III: Question 4 or 5	**18**
6 & 7	Unit - IV: Question 6 or 7	**18**
8 & 9	Unit - V: Question 8 or 9	**18**
	TOTAL	**100**

Semester - III

National Service Scheme
Category: AE Course
(All programs)
(Practical)

Course Code	**:**	HS238LA	**CIE**	**:**	50 Marks
Credit distribution L:T:P	**:**	0:0:1	**SEE**	**:**	50 Marks
Pedagogical Hours (L:T:P:S)	**:**	0:0:30:15	**CIE Duration**	**:**	1 Hours
Total credits:	**:**	1	**SEE Duration**	**:**	2 Hours

Pre requisites:

1. Students should have service oriented mindset and social concern.

2. Students should have dedication to work at any remote place, any time with available resources and proper time management for the other works.

3. Students should be ready to sacrifice some of the timely will and wishes to achieve service oriented targets on time.

1. **Content**

Students has to take up any one activities on below mentioned topics and has to prepare contents for awareness and technical contents for implementation of the projects and has to present strategies for implementation of the same. Compulsorily has to attend one camp. CIE will be evaluated based on their presentation, approach and implementation strategies. (Any one of the below mentioned activity)

1. Helping local schools to achieve good result and enhance their enrolment in Higher/technical/ vocational education.

2. Preparing an actionable business proposal for enhancing the village/ farmer income and approach for implementation.

3. Developing Sustainable Water management system for rural/ urban areas and implementation approaches.

4. Setting of the information imparting club for women leading to contribution in social and economic issues.

5. Spreading public awareness/ government schemes under rural outreach program. (minimum 5 programs)

6. Contribution to any national level initiative of Government of India. For eg. Digital India, Skill India, Swachh Bharat, Atmanirbhar Bharath, Make in India, Mudra scheme, Skill development programs etc

7. Social connect and responsibilities

8. Plantation and adoption of plants. Know your plants

9. Organic farming , Indian Agriculture (Past, Present and Future) Connectivity for marketing

10. Waste management – Public, Private and Govt organization, 5 R's

11. Water conservation techniques – Role of different stakeholders - Implementation

12. Govt . School Rejuvenation and assistance to achieve good infrastructure.

13. Organize National integration and social harmony events/ workshops/ seminars. (Minimum 2 programs)

2. AND ONE NSS - CAMP

Course Outcomes: After completing the course, the students will be able to:

CO1: Understand the importance of his/her responsibilities towards society.

CO2: Analyze the environmental and societal problems/ issues and will be able to design solutions for the same.

CO3: Evaluate the existing system and to propose practical solutions for the same for sustainable development.

CO4: Implement government or self-driven projects effectively in the field.

RUBRICS FOR THE CONTINUOUS INTERNAL EVALUATION		
#	**COMPONENTS**	**MARKS**
1	**Presentation 1** Selection of topic- (phase 1) Justification for Importance, need of the hour with surveyed data	**10**
2	**Presentation 2** (phase 2) Content development, strategies for implementation methodologies.	**10**
3	Case Study-based Teaching-Learning	**10**
4	Sector wise study and consolidation	**10**
5	Video based presentation (4-5 minutes per student)	**10**
	MAXIMUM MARKS FOR THE CIE	**50**

RUBRICS FOR SEMESTER END EXAMINATION (LAB)		
Q.NO.	**CONTENTS**	**MARKS**
1	Implementation strategies of the project with report	**50**
	TOTAL	**50**

Semester - III					
National Cadet Corps Category: AE Course (All programs) (Practical)					
Course Code	**:**	HS238LB	**CIE**	**:**	50 Marks
Credit distribution L:T:P	**:**	0:0:1	**SEE**	**:**	50 Marks
Pedagogical Hours (L:T:P:S)	**:**	0:0:30:15	**CIE Duration**	**:**	1 Hours
Total credits:	**:**	1	**SEE Duration**	**:**	2 Hours

Unit-I

Introduction to NCC and Indian Armed Forces – Introduction to NCC, Aims, organization, and significance of NCC, Overview of the Indian Armed Forces, Organizational structure, Ranks, Entry into the Indian Army.

Unit-II

Drill- Foot Drill, Drill ki Aam Hidayaten, Word ki Command, Savdhan, Vishram, Aram Se, Murdna, KadvarSizing, Teen Line Banana, Khuli Line, Nikat Line, Khade Khade Salute Karna.

Unit-III

Weapon Training, Trekking, Map Reading and JD- Overview of weapons, Introduction to SLR (Self Loading Rifle), Introduction to trekking, Types of maps, Finding North, Determining own position, Knots, Judging distance

Unit-IV

Social Service and Disaster Management- Social Service and Community Development, Disaster management, Assistance during natural disasters, Fire prevention, Fire-fighting equipment.

Course Outcomes: After completing the course, the students will be able to:

CO1: Understand the aims, organization, and significance of the National Cadet Corps (NCC) and gain a basic overview of the Indian Armed Forces, including their structure, ranks, and entry options.

CO2: Demonstrate proficiency in foot drill commands and movements, thereby developing discipline, coordination, and the ability to function effectively as part of a group.

CO3: Understand the fundamentals of weapon training, trekking, map reading, and judging distance, including safe handling of the Self-Loading Rifle (SLR) and basic fieldcraft skills essential to the army way of life.

CO4: Apply knowledge of social service, community development, and disaster management by participating in activities related to public welfare, disaster response, fire prevention, and emergency assistance.

References

[1] R. K. Gupta, *NCC Cadet Hand Book*. New Delhi: Ramesh Publishing House, Book Code: R-1991, HSN Code: 49011010, ISBN: 978-93-87918-57-3

[2] *National cadet corps (NCC) india official website*, `https://nccindia.ac.in`. [Online]. Available: `https://nccindia.ac.in`

RUBRICS FOR THE CONTINUOUS INTERNAL EVALUATION		
#	COMPONENTS	MARKS
1	**Presentation 1** Selection of topic- (phase 1) Justification for Importance, need of the hour with surveyed data	**10**
2	**Presentation 2** (phase 2) Content development, strategies for implementation methodologies.	**10**
3	Case Study-based Teaching-Learning	**10**
4	Sector wise study and consolidation	**10**
5	Video based presentation (4-5 minutes per student)	**10**
	MAXIMUM MARKS FOR THE CIE	**50**

RUBRICS FOR SEMESTER END EXAMINATION (LAB)		
Q.NO.	CONTENTS	MARKS
1	Implementation strategies of the project with report	**50**
	TOTAL	**50**

Semester - III

Physical Education: Sports and Athletics
Category: AE Course
(All programs)
(Practical)

Course Code	**:**	HS238LC	**CIE**	**:**	50 Marks
Credit distribution L:T:P	**:**	0:0:1	**SEE**	**:**	50 Marks
Pedagogical Hours (L:T:P:S)	**:**	0:0:30:15	**CIE Duration**	**:**	1 Hours
Total credits:	**:**	1	**SEE Duration**	**:**	2 Hours

Introduction of Physical Education and Sports

General, Specific and Conditioning warm up exercises Sports Engineering Any 1 Major Game Intramural Competitions

1. **Kho-Kho-** Giving Kho, Single chain, Pole dive, Pole turning, 3-6 Up
2. **Throwball-** Service,Receive,Spin pass, Simple pass, Jump throw
3. **Netball-** Step with ball, Shooting, Passing, Blocking
4. **Ball Badminton-** Service, Fore hand receive, Back hand receive, Spin smash, Rally
5. **Volleyball-** Attack, Block, Service, Upper hand pass, Lower hand pass
6. **Handball-** Step with ball, Shooting, Passing,Blocking, Dribbling
7. **Football-** Dribbling, Chest Drop, Ball Control,Thigh Drop, Shooting

Topics for Viva:

1. On rules and regulations pertaining to the games/sports.
2. On the dimensions of the court, size/weight of the ball, and standards pertaining to the respective sport/game.
3. Popular players and legends at the State, National, and International levels.
4. Recent events, including winners and runners-up in the respective sport/game.
5. General awareness about the sport/game and sports-related activities on the college campus.

Course Outcomes: After completing the course, the students will be able to:

CO1: Understand the basic principles and practices of Physical Education and Sports.

CO2: Instruct the Physical Activities and Sports practices for Healthy Living

CO3: To develop professionalism among students to conduct, organize and Officiate Physical Education and Sports events at schools and community level

CO4: Understand and Develop Sports and Engineering.

References

[1] J. P. Muller, *Health, Exercise and Fitness*. Delhi: Sports, 2000

[2] A. Vanaik, *Play Field Manual*. New Delhi: Friends Publication, 2005

[3] IAAF, *IAAF manual*

[4] M. J. Vishwanath, *Track and Field Marking and Athletics Officiating Manual*. Shimoga: Silver Star Publication, 2002

[5] S. Oldenburg, *Complete Conditioning for Volleyball*. Human Kinetics, 2015

RUBRICS FOR THE CONTINUOUS INTERNAL EVALUATION		
#	COMPONENTS	MARKS
1	Attendance	**10**
2	Sports Engineering	**20**
3	Court Measurement (Marking) operation and skill	**20**
	MAXIMUM MARKS FOR THE CIE	**50**

RUBRICS FOR SEMESTER END EXAMINATION (LAB)		
Q.NO.	CONTENTS	MARKS
1	Demonstration /Display and Presentation / Skill tests	**40**
1	Viva Voce	**10**
	TOTAL	**50**

Semester - III					
Music **Category: AE Course** **(All programs)** **(Practical)**					
Course Code	**:**	HS238LD	**CIE**	**:**	50 Marks
Credit distribution L:T:P	**:**	0:0:1	**SEE**	**:**	50 Marks
Pedagogical Hours (L:T:P:S)	**:**	0:0:30:15	**CIE Duration**	**:**	1 Hours
Total credits:	**:**	1	**SEE Duration**	**:**	2 Hours

Content

1. Introduction to different genres of music
2. Evolution of genres in India: Inspiration from the world
3. Ragas, time and their moods in Indian Classical Music
4. Identification of ragas and application into contemporary songs
5. Adding your touch to a composition
6. Maths and Music: A demonstration
7. Harmonies in music
8. Chords: Basics and application into any song
9. Music Production-I
10. Music Production-II

Course Outcomes: After completing the course, the students will be able to:

CO1: Understand basics of Music and improve their skills.

CO2: Appreciate the impact on health and well-being.

CO3: Perform and present music in a presentable manner.

CO4: Develop skills like team building and collaboration..

References

[1] S. R. Janakiraman, *Ragas in Indian Music*

[2] G. E. Ruckert, *Music in North India: Experiencing Music, Expressing Culture*

[3] B. Owsinski, *The Mixing Engineer's Handbook*

[4] R. Iyer, *Elements Of Indian Music: The Melakarta System*. 2018, ISBN: 1513460854

RUBRICS FOR THE CONTINUOUS INTERNAL EVALUATION		
#	**COMPONENTS**	**MARKS**
1	Solo Performance (Vocal/Instrumental)	**20**
2	Group Performance	**10**
3	Music Making using a Digital Audio Workstation (DAW)	**10**
4	Sound Synthesis using Vital VSTi	**10**
	MAXIMUM MARKS FOR THE CIE	**50**

RUBRICS FOR SEMESTER END EXAMINATION (LAB)		
Q.NO.	**CONTENTS**	**MARKS**
1	Solo Performance (Vocal/Instrumental)	**20**
2	Group Performance	**10**
3	Report	**20**
	TOTAL	**50**

Semester - III					
Dance **Category: AE Course** **(All programs)** **(Practical)**					
Course Code	**:**	HS238LE	**CIE**	**:**	50 Marks
Credit distribution L:T:P	**:**	0:0:1	**SEE**	**:**	50 Marks
Pedagogical Hours (L:T:P:S)	**:**	0:0:30:15	**CIE Duration**	**:**	1 Hours
Total credits:	**:**	1	**SEE Duration**	**:**	2 Hours

Unit-I

Foundations of Dance and Music:

- Introduction to dance as an art form and medium of expression
- Role of dance in physical, mental, and emotional well-being
- Body awareness: posture, alignment, flexibility, and balance
- Warm-up, stretching, and cool-down techniques for dancers
- Understanding music fundamentals: beats, rhythm, tempo, and counts
- Coordination of basic movements with music

Unit-II

Dance Styles and Skill Development:

- Overview of major dance forms: Classical, Folk/Eastern, and Western
- Basic steps, footwork, and expressions in different dance styles
- Cultural and stylistic features of various dance forms
- Assessment of student interest and aptitude through interaction
- Grouping of students based on selected dance styles
- Style-specific movement practice and improvisation

Unit-III

Choreography and Performance:

- Advanced techniques in selected dance styles
- Expert sessions and guest workshops in respective dance forms
- Components of performance: space, energy, timing, and expression
- Introduction to choreographies and structured dance routines
- Basics of choreography: formations, transitions, and storytelling
- Creation and performance of solo or group choreography

ASSESSMENT AND EVALUATION PATTERN

Continuous Internal Evaluation (CIE)

- CIE is carried out through solo and group dance performances. Solo Dance Performance: Students are required to choreograph and perform a 2–3-minute routine. The performance is evaluated based on dance technique, expressions, and choreography. Group Dance Performance: A group comprising of 5-6 members will be made by the instructor. The group performance is evaluated based on coordination, technique, and choreography.

Experiential Learning (EL):

- Students will engage in experiential learning activities to explore the cultural, historical, and social dimensions of dance. Possible experiential learning topics include:
- Dance postures and movements depicted in temple carvings
- Temples historically associated with dance traditions
- Ritualistic and ceremonial dances performed in temples
- Regional folk dance forms and their cultural significance
- Dance as a medium of storytelling in religious and social contexts
- Costumes, jewellery, and aesthetics of temple and folk dances
- Community based folk dances and festival performances
- Role of music in traditional dance practices
- Documentation of local dance heritage
- Interaction with practitioners of folk or temple dance traditions

References

[1] J. M. Smith-Autard, *Dance Composition: A Practical Guide to Creative Success in Dance Making*, 6th. Bloomsbury Academic / Methuen Drama, 2010, ISBN: 978-1408115640

[2] P. A. Sofras, *Dance Composition Basics*, 2nd. Human Kinetics, 2019, ISBN: 978-1492571254

[3] L. A. Blom and L. T. Chaplin, *The Intimate Act of Choreography*, 1st. University of Pittsburgh Press, 1982, ISBN: 978-0822953425

RUBRICS FOR THE CONTINUOUS INTERNAL EVALUATION		
#	**COMPONENTS**	**MARKS**
1	Solo Performance	**20**
2	Group Performance	**10**
3	Experiential Learning	**20**
	MAXIMUM MARKS FOR THE CIE	**50**

RUBRICS FOR SEMESTER END EXAMINATION (LAB)		
Q.NO.	**CONTENTS**	**MARKS**
1	Solo Performance	**20**
2	Group Performance	**10**
3	Report	**20**

RUBRICS FOR SEMESTER END EXAMINATION (LAB) (Continued)		
Q.NO.	**CONTENTS**	**MARKS**
	TOTAL	**50**

Semester - III

Theatre (Light Camera and Action)
Category: AE Course
(All programs)
(Practical)

Course Code	**:**	HS238LF	**CIE**	**:**	50 Marks
Credit distribution L:T:P	**:**	0:0:1	**SEE**	**:**	50 Marks
Pedagogical Hours (L:T:P:S)	**:**	0:0:30:15	**CIE Duration**	**:**	1 Hours
Total credits:	**:**	1	**SEE Duration**	**:**	2 Hours

Prerequisites:

1. Students should have creative oriented mindset and social concern. 2. Students should have dedication to work with their classmates for long hours until a collective goal is reached. 3. Students should be ready to sacrifice some of the timely will and wishes to achieve targets on time.

Content

1. Break the Ice
2. Introduction to freedom Talk to each and every single person for a period of 5 complete minutes. This is aimed at to make everyone in the room comfortable with each other. This helps everyone get over social anxiety, Shyness and Nervousness.
3. Ura
4. Rhythm Voice Projection, Voice Modulation, Weeping and Coughing Voice projection is the strength of speaking or singing whereby the voice is used powerfully and clearly. It is a technique employed to command respect and attention, as when a teacher talks to a class, or simply to be heard clearly, as used by an actor in a theatre.
5. Its Leviosa, Not Leviosaaa
6. Speech work: Diction, Intonation, Emphasis, Pauses, Pitch and Volume Tempo Dialogues delivery. The art of dialogue delivery plays a vital role in in ensuring the efficacy of communication especially from the dramatic aspect of it, this unit discusses some tips to help the young actors improve their dialogue delivery skills:
7. Elementary, My dear Watson.
8. Responsibilities of an actor tools of an actor character analysis Observations aspects, Stage presence, concentration, conviction, confidence, energy and directionality.
9. Show Time
10. Pick a genre: COMEDY, THRILLER, HORROR, and TRAGEDY: Showcase a performance. Stylized acting with reference to historical and mythological plays. Mime: conventional, occupational and pantomime Mono acting: different types of characters

CIE's will be evaluated through mono-acting or dialogue. The students need to use whatever they've learnt through the course of the drama class. Judges/Teachers can award the marks accordingly. Certificates won outside of college, can be submitted for evaluation as well.

For SEE's. Students need to form groups of 4-6. They need to pick a genre and enact a play of at least 20 mins long. The venue will be IEM auditorium. No mics should be used. They will be given 2 weeks to prepare.

Course Outcomes: After completing the course, the students will be able to:

CO1: Develop a range of Theatrical Skills and apply them to create a performance.

CO2: Work collaboratively to generate, develop and communicate ideas.

CO3: Develop as creative, effective, independent and reflective students who are able to make inform ed choices in process and performance.

CO4: Develop an awareness and understanding of the roles and processes undertaken in contemporary professional theatre practice.
References
[1] P. Brook, *The Empty Space*
[2] A. Bogart and T. Landau, *A Practical Guide to Viewpoints and Composition*

RUBRICS FOR THE CONTINUOUS INTERNAL EVALUATION		
#	**COMPONENTS**	**MARKS**
1	**Presentation 1** Selection of topic- (phase 1) Justification for Importance, need of the hour with surveyed data	**10**
2	**Presentation 2** (phase 2) Content development, strategies for implementation methodologies.	**10**
3	Case Study-based Teaching-Learning	**10**
4	Sector wise study and consolidation	**10**
5	Video based presentation (4-5 minutes per student)	**10**
	MAXIMUM MARKS FOR THE CIE	**50**

RUBRICS FOR SEMESTER END EXAMINATION (LAB)		
Q.NO.	**CONTENTS**	**MARKS**
1	Implementation strategies of the project with report	**50**
	TOTAL	**50**

Semester - III					
Art Work and Painting **Category: AE Course** **(All programs)** **(Practical)**					
Course Code	**:**	HS238LG	**CIE**	**:**	50 Marks
Credit distribution L:T:P	**:**	0:0:1	**SEE**	**:**	50 Marks
Pedagogical Hours (L:T:P:S)	**:**	0:0:30:15	**CIE Duration**	**:**	1 Hours
Total credits:	**:**	1	**SEE Duration**	**:**	2 Hours

Prerequisites

Although there are no prerequisite qualifications for this subject, students must have a basic understanding of and interest in the fields of art and design in order to enroll in it.

Innovative Experiments

1. Use points, line and curves to create various shapes and forms
2. Use of shapes and forms to create various objects and structures
3. Recognizing distinctions in objects when viewed from various perspectives and grasping basic notions of perspective
4. Students will be introduced to the significance of color in art, as well as the principles of color theory and application.
5. Applied the concepts of unity, harmony, balance, rhythm, emphasis and proportion, abstraction and stylization to create a composition.
6. Learn how to use which materials and for what types of art and textures.
7. Use of the above concepts to create art through the medium of collage, mosaic, painting, mural, batik, tie and dye.
8. Real world application of the above concepts in the form of book cover design and illustration, cartoon, poster, advertisements, magazine, computer graphics and animation
9. Familiarization with the many art forms and techniques of expression found throughout India.

AND ONE EDUCATIONAL VISIT TO AN ART MUSEUM / INSTITUTE / GALLERY Students must turn in assignments for each of the above said topics on a weekly basis and have to compulsorily take part in the museum visit. CIE will be evaluated based on a still life piece, a composition using any one of the media of composition and a presentation on Indian art styles and creation of a piece pertaining to the presented art style.

Course Outcomes: After completing the course, the students will be able to:

CO1: To use lines, shapes, and colors to depict the various sentiments and moods of life and nature.

CO2: To use one's creativity to develop forms and color schemes, as well as the ability to portray them effectively in drawing and painting on paper.

CO3: To develop the ability to properly use drawing and painting materials (surfaces, tools and equipment, and so on).

CO4: To improve their observation abilities by studying everyday items as well as numerous geometrical and non- geometrical (i.e. organic) shapes found in life and nature and to hone their drawing and painting talents in response to these insights.

References

[1] D. Lynch, *Catching the Big Fish: Meditation, Consciousness, and Creativity*

[2] D. Bayles and T. Orland, *Art and Fear: Observations on the Perils (and Rewards) of Artmaking*

RUBRICS FOR THE CONTINUOUS INTERNAL EVALUATION		
#	COMPONENTS	MARKS
1	**Presentation 1** Selection of topic- (phase 1) Justification for Importance, need of the hour with surveyed data	**10**
2	**Presentation 2** (phase 2) Content development, strategies for implementation methodologies.	**10**
3	Case Study-based Teaching-Learning	**10**
4	Sector wise study and consolidation	**10**
5	Video based presentation (4-5 minutes per student)	**10**
	MAXIMUM MARKS FOR THE CIE	**50**

RUBRICS FOR SEMESTER END EXAMINATION (LAB)		
Q.NO.	CONTENTS	MARKS
1	Implementation strategies of the project with report	**50**
	TOTAL	**50**

Semester - III					
Photography and Film Making **Category: AE Course** **(All programs)** **(Practical)**					
Course Code	**:**	HS238LH	**CIE**	**:**	50 Marks
Credit distribution L:T:P	**:**	0:0:1	**SEE**	**:**	50 Marks
Pedagogical Hours (L:T:P:S)	**:**	0:0:30:15	**CIE Duration**	**:**	1 Hours
Total credits:	**:**	1	**SEE Duration**	**:**	2 Hours

Contents

1. Introduction to photography.
2. Understanding the terminologies of DSLR.
3. Elements of photography.
4. Introduction to script writing, storyboarding.
5. Understanding the visualization and designing a set.
6. Basics of film acting
7. Video editing using software
8. Introduction to cinematography.
9. Understanding about lighting and camera angles.
10. Shooting a short film.

Students must form groups of 2-4 and present a short film which shall be given by the experts. The experts shall judge the groups and award marks for the same. CIE will be evaluated based on their presentation, approach and implementation strategies. Students need to submit their certificates of any event they participated or bagged prizes in. This shall also be considered for CIE evaluation.

11. For EL experiment

Course Outcomes: After completing the course, the students will be able to:

CO1: Understand basics of photography and videography and improve their skills.

CO2: Appreciate the skills acquired from photography.

CO3: Perform and present photos and films in a presentable manner.

CO4: Develop skills like team building and collaboration.

References

[1] H. Carroll, *Read This If You Want to Take Great Photographs*

[2] S. Kelby, *The Digital Photography Book: Part 1*

RUBRICS FOR THE CONTINUOUS INTERNAL EVALUATION		
#	**COMPONENTS**	**MARKS**
1	**Presentation 1** Selection of topic- (phase 1) Justification for Importance, need of the hour with surveyed data	**10**
2	**Presentation 2** (phase 2) Content development, strategies for implementation methodologies.	**10**
3	Case Study-based Teaching-Learning	**10**

RUBRICS FOR THE CONTINUOUS INTERNAL EVALUATION (Continued)		
#	COMPONENTS	MARKS
4	Sector wise study and consolidation	**10**
5	Video based presentation (4-5 minutes per student)	**10**
	MAXIMUM MARKS FOR THE CIE	**50**

RUBRICS FOR SEMESTER END EXAMINATION (LAB)		
Q.NO.	CONTENTS	MARKS
1	Implementation strategies of the project with report	**50**
	TOTAL	**50**

Semester - III					
Bridge Course Mathematics **Category: Bridge Course** **(Common to All Programs)** **(Theory)**					
Course Code	**:**	MAT139BT	**CIE**	**:**	50 Marks
Credit distribution L:T:P	**:**	2:0:0	**SEE**	**:**	—
Pedagogical Hours (L:T:P:S)	**:**	30:0:0:30	**CIE Duration**	**:**	1.5 Hours
Total credits:	**:**	2	**SEE Duration**	**:**	—

Unit - I	**10 Hours**

Multivariable Calculus: Partial Differentiation: Introduction, simple problems. Total derivative, composite functions. Jacobians – simple problems. **Vector Differentiation:** Introduction, velocity and acceleration, gradient, divergence – solenoidal vector function, curl – irrotational vector function and Laplacian, simple problems.

Unit - II	**10 Hours**

Differential Equations: Higher order linear differential equations with constant coefficients, solution of homogeneous equations - Complementary functions. Non-homogeneous equations – Inverse differential operator method of finding particular integral based on input function (force function).

Unit - III	**10 Hours**

Numerical Methods:Solution of algebraic and transcendental equations – Intermediate value property, Secant method, Newton-Raphson method. Solution of first order ordinary differential equations – Taylor series and 4th order Runge-Kutta methods. Numerical integration – Simpson's 1/3rd, 3/8th and Weddle's rules. (All methods without proof).

Course Outcomes: After completing the course, the students will be able to:

CO1: Explain and outline the fundamental concepts of partial differentiation, vector calculus, higher order linear differential equations and basic numerical methods used in engineering mathematics.

CO2: Apply and compute solutions using multivariable calculus, vector differentiation, differential equation techniques, numerical algorithms to solve elementary engineering and scientific problems.

CO3: Analyze and evaluate solutions obtained through differential calculus, vector operations, differential equations, numerical methods to assess their accuracy and applicability in practical situations.

CO4: Integrate and summarize the knowledge of calculus, vector methods, differential equations, numerical techniques to build mathematical competence and support life-long learning in engineering studies.

References

[1] B. S. Grewal, *Higher Engineering Mathematics*, 44th ed. Khanna Publishers, 2015, ISBN: 9788193328491

[2] B. V. Ramana, *Higher Engineering Mathematics*, 11th ed. Tata McGraw-Hill Education, 2010, ISBN: 9780070634190

[3] N. P. Bali and M. Goyal, *A Textbook of Engineering Mathematics*, 7th ed. Laxmi Publications, 2010, ISBN: 9788131808320

RUBRICS FOR THE CONTINUOUS INTERNAL EVALUATION		
#	**COMPONENTS**	**MARKS**
1	**QUIZZES:** Quizzes will be conducted in online/offline mode. TWO quizzes will be conducted and Each Quiz will be evaluated for 10 Marks. **The sum of the two will be the final quiz marks.**	**20**
2	**TESTS:** Students will be evaluated in test through descriptive questions with different complexity levels (Revised Bloom's Taxonomy Levels: Remembering, Understanding, Applying, Analyzing, Evaluating, and Creating). TWO tests will be conducted. Each test will be evaluated for 30 Marks, adding upto 60 Marks. **Final test marks will be average of two tests.**	**30**
	MAXIMUM MARKS FOR THE CIE	**50**

Semester - IV

Probability and Applied Statistics
Category: Professional Core Course
(Theory)

Course Code	**:**	IM341AT	**CIE**	**:**	100 Marks
Credit distribution L:T:P	**:**	3:0:0	**SEE**	**:**	100 Marks
Pedagogical Hours (L:T:P:S)	**:**	45:0:0:45	**CIE Duration**	**:**	2 Hours
Total credits:	**:**	3	**SEE Duration**	**:**	3 Hours

Unit - I **9 Hours**

Introduction to Probability: Sample spaces and Events, Interpretations of probability, Addition rules, Conditional probability Multiplication and Total probability rules, Independence, Bayes Theorem, Numerical Problems **Concepts of Random Variables:** Random Variables, Discrete and continuous random variables. Probability distributions and mass functions, Probability Density functions, Numerical Problems

Unit - II **9 Hours**

Discrete Probability Distributions: Discrete Uniform, Binominal, Poisson, Geometric, Negative binomial, Applications, Numerical Problems. Implementation using MATLAB.

Unit - III **9 Hours**

Continuous Probability Distributions: Continuous uniform, Normal, Normal approximations to Binomial and Poisson Distributions, Exponential Distributions, Weibull Distributions, Applications, Numerical Problems, Implementation using MATLAB.

Unit - IV **9 Hours**

Simple and Multiple Linear Regression: Empirical models, Simple Linear Regression, Properties of Least square Estimators, Multiple Linear Regression model, Numerical Problems.

Unit - V **9 Hours**

Statistical Inference for a single sample: Test of hypothesis - Null and alternative hypothesis, Procedure for statistical testing, Type I and Type II errors, level of significance, Tests involving the normal distribution, one–tailed and two–tailed tests, P–value, Inference on the mean of a normal population (variance known and unknown), Inference on the variance of a Normal population, Testing for Goodness of Fit, Numerical Problems

Course Outcomes: After completing the course, the students will be able to:

CO1: Describe and report data set using data analysis, presentation and interpretation techniques to understand various phenomena in the fields of science and engineering.

CO2: Apply various statistical processing techniques to handle a set of data to estimate probabilities.

CO3: Apply an appropriate statistical tool and analyze a specific set of data to estimate and draw conclusions about population parameters

CO4: Draw inferences about population parameters and relations between variables based on analysis of sample data

References

[1] D. C. Montgomery and G. C. Runger, *Applied Statistics and Probability for Engineers*, 4th ed. Wiley, 2007, Asia Student Edition, ISBN: 9788126523153

[2] R. I. Levin and D. S. Rubin, *Statistics for Management*, 7th ed. Prentice Hall India, 1997, ISBN: 9780134762920

[3] R. E. Walpole et al., *Probability and Statistics for Engineers and Scientists*, 8th ed. Pearson Education, 2007, ISBN: 9788131715529

[4] *Microsoft excel*, Statistical Analysis Software

[5] *Minitab*, Statistical Analysis Software

[6] *Matlab*, Numerical Computing Software

[7] *R statistical computing environment*, Open-source Statistical Software

[8] NPTEL, *Introduction to data analytics*, `http://172.16.44.44/nnptel.html`, Course sponsored by Aricent

[9] NPTEL, *Introduction to probability and statistics*, `https://onlinecourses.nptel.ac.in/noc22_mg31/preview`, 2022

[10] Pennsylvania State University, *Statistics program – undergraduate studies*, `https://newonlinecourses.science.psu.edu/statprogram/undergraduate-studies`

[11] Khan Academy, *Statistics and probability*, `https://www.khanacademy.org/math/statistics-probability`

RUBRICS FOR THE CONTINUOUS INTERNAL EVALUATION		
#	**COMPONENTS**	**MARKS**
1	**QUIZZES:** Quizzes will be conducted in online/offline mode. TWO quizzes will be conducted and each Quiz will be evaluated for 10 Marks. **The sum of the two will be the final quiz marks.**	**20**
2	**TESTS:** Students will be evaluated in test through descriptive questions with different complexity levels (Revised Bloom's Taxonomy Levels: Remembering, Understanding, Applying, Analyzing, Evaluating, and Creating). THREE tests will be conducted (Two regular tests and One optional Improvement test). Each test will be evaluated for 50 Marks, adding upto 100 Marks. **Final test marks will be reduced to 40**.	**40**
3	**EXPERIENTIAL LEARNING:** Students shall be evaluated based on the creativity, innovation, and practical implementation of their proposed solutions. The evaluation shall be conducted in two phases. The implementation modalities for experiential learning activity shall be decided considering the nature of the course and its interdisciplinary requirements. **Each phase shall carry 20 marks, with the total evaluation carrying 40 marks.**	**40**
	MAXIMUM MARKS FOR THE CIE	**100**

RUBRICS FOR SEMESTER END EXAMINATION (THEORY)		
Q.NO.	**CONTENTS**	**MARKS**
	PART A	
1	Objective type questions covering entire syllabus	**10**
	PART B (Maximum of THREE Sub-divisions only)	
2	Unit - I: Compulsory	**18**
3	Unit - II: Compulsory	**18**
4 & 5	Unit - III: Question 4 or 5	**18**
6 & 7	Unit - IV: Question 6 or 7	**18**
8 & 9	Unit - V: Question 8 or 9	**18**

RUBRICS FOR SEMESTER END EXAMINATION (THEORY) (Continued)		
Q.NO.	**CONTENTS**	**MARKS**
	TOTAL	**100**

Semester - IV					
Material Science for Engineers **Category: Professional Core Course** **(Theory)**					
Course Code	**:**	ME242TB	**CIE**	**:**	100 Marks
Credit distribution L:T:P	**:**	3:0:0	**SEE**	**:**	100 Marks
Pedagogical Hours (L:T:P:S)	**:**	45:0:0:45	**CIE Duration**	**:**	2 Hours
Total credits:	**:**	3	**SEE Duration**	**:**	3 Hours

Unit - I	**7 Hours**
Review of mechanical properties: Stress-strain diagram, elastic deformation, plastic deformation, Yield point phenomena, hardness, impact energy, ductile fracture, brittle fracture. **Fatigue and Creep of materials:** Types of cyclic loading, S–N curve and fatigue properties, mechanism of fatigue failure, Factors affecting fatigue life, Creep curve and stages, Creep mechanisms.	
Unit - II	**8 Hours**
Crystal imperfections and plastic deformation: Types of defects, dislocations in plastic deformation, Solid solutions, Hume Rothery Rules, Critical resolved shear stress. **Strengthening mechanisms:** Grain size strengthening, Solid solution strengthening, Strain (work) hardening, Precipitation and dispersion strengthening.	
Unit - III	**9 Hours**
Phase Diagrams and Fe-C equilibrium diagram: Introduction to phase, phase diagrams, Gibbs phase rule, Lever rule, cooling curves, Isomorphous alloy system, Iron-Iron carbide phase diagram-Invariant reactions. **Solidification:** Nucleation, homogeneous nucleation, heterogeneous nucleation, crystal growth.	
Unit - IV	**9 Hours**
Heat treatment of ferrous materials: Annealing, normalizing, hardening, tempering. TTT and CCT Diagrams. **Surface heat treatment processes:** carburizing, nitriding, cyaniding, flame, and induction hardening. Defects in heat treatment.	
Unit - V	**7 Hours**
Modern Semiconductor Processing: Introduction, Unit processes - oxidation, photolithography, etching and doping, process sequences, information flow, process organization, goals of semiconductor manufacturing. **Heat treatment of Electronic Materials:** Purpose of heat treatment in electronic materials, Annealing, baking, solder reflow, oxidation, diffusion, rapid thermal processing.	

Course Outcomes: After completing the course, the students will be able to:

CO1: Understand the mechanical properties and deformation mechanisms of engineering materials.

CO2: Apply stress-strain concepts to evaluate material behavior.

CO3: Analyze the effects of plastic deformation, creep, fatigue, and fracture on material performance.

CO4: Evaluate and predict material behavior under different mechanical and thermal conditions relevant to structural and semiconductor applications.

References

[1] V. Raghavan, *Materials Science and Engineering: A First Course*, 5th ed. PHI Learning Pvt. Ltd., 2015, ISBN: 9788120350922

[2] W. D. C. Jr. and D. G. Rethwisch, *Materials Science and Engineering: An Introduction*, 10th ed. John Wiley & Sons, 2018, ISBN: 9781119453565

[3] B. S. N. Shetty, *Engineering Materials*. McGraw-Hill Education, 2017, ISBN: 9780070581940

[4] J. W. Mayer and S. S. Lau, *Electronic Materials Science: For Integrated Circuits in Si and GaAs*, 1st ed. Macmillan Publishing Company, 1989, ISBN: 9780023781407

[5] G. S. May and C. J. Spanos, *Fundamentals of Semiconductor Manufacturing and Process Control*. John Wiley & Sons, 2000, ISBN: 9780471784067

[6] NPTEL, *Fundamentals of electronic materials and devices*, https://nptel.ac.in/courses/113106065

[7] NPTEL, *Fundamentals of electronic device fabrication*, https://nptel.ac.in/courses/113106062

[8] NPTEL, *Basics of materials engineering*, https://nptel.ac.in/courses/112106293

[9] NPTEL, *Introduction to materials science and engineering*, https://nptel.ac.in/courses/113102080

RUBRICS FOR THE CONTINUOUS INTERNAL EVALUATION		
#	**COMPONENTS**	**MARKS**
1	**QUIZZES:** Quizzes will be conducted in online/offline mode. TWO quizzes will be conducted and each Quiz will be evaluated for 10 Marks. **The sum of the two will be the final quiz marks.**	**20**
2	**TESTS:** Students will be evaluated in test through descriptive questions with different complexity levels (Revised Bloom's Taxonomy Levels: Remembering, Understanding, Applying, Analyzing, Evaluating, and Creating). THREE tests will be conducted (Two regular tests and One optional Improvement test). Each test will be evaluated for 50 Marks, adding upto 100 Marks. **Final test marks will be reduced to 40**.	**40**
3	**EXPERIENTIAL LEARNING:** Students shall be evaluated based on the creativity, innovation, and practical implementation of their proposed solutions. The evaluation shall be conducted in two phases. The implementation modalities for experiential learning activity shall be decided considering the nature of the course and its interdisciplinary requirements. **Each phase shall carry 20 marks, with the total evaluation carrying 40 marks.**	**40**
	MAXIMUM MARKS FOR THE CIE	**100**

RUBRICS FOR SEMESTER END EXAMINATION (THEORY)		
Q.NO.	**CONTENTS**	**MARKS**
	PART A	
1	Objective type questions covering entire syllabus	**10**
	PART B (Maximum of THREE Sub-divisions only)	
2	Unit - I: Compulsory	**18**
3	Unit - II: Compulsory	**18**
4 & 5	Unit - III: Question 4 or 5	**18**
6 & 7	Unit - IV: Question 6 or 7	**18**
8 & 9	Unit - V: Question 8 or 9	**18**
	TOTAL	**100**

Semester - IV					
Principles of Ergonomics **Category: Professional Core Course** **(Theory & Practical)**					
Course Code	**:**	IM243AI	**CIE**	**:**	100+ 50 Marks
Credit distribution L:T:P	**:**	3:0:1	**SEE**	**:**	100+ 50 Marks
Pedagogical Hours (L:T:P:S)	**:**	45:0:30:45	**CIE Duration**	**:**	2+2 Hours
Total credits:	**:**	4	**SEE Duration**	**:**	3+3 Hours

Unit - I	**9 Hours**
Foundations of Ergonomics: Systems approach in ergonomics, Ergonomics in product design, workplace design and service systems, Ergonomic design process, Ergonomics standards and guidelines, Introduction to human performance, Introduction to IoT and ergonomics; IoT architecture and components; sensors and actuators for ergonomic applications	
Unit - II	**9 Hours**
Anthropometry and Workspace Design: Human body dimensions and variability, Static and dynamic anthropometry, Anthropometric data and percentiles, Design for extremes and design for adjustability, Posture and body mechanics, Principles of workstation design, Design of seated and standing workplaces, Reach, clearance and field of vision, Layout of controls and work surfaces	
Unit - III	**9 Hours**
Human–Machine Interaction and Ergonomic Applications: Displays and controls, Principles of display design, Control types and selection, Human-computer interaction, User-centred design, Ergonomic evaluation methods, Checklists and ergonomic audits, Workplace assessment, Case studies in industrial and office ergonomics	
Unit - IV	**9 Hours**
Environmental and Cognitive Ergonomics: Physical work environment, Lighting and illumination, Noise and vibration, Thermal comfort, Effects of environment on performance, Cognitive ergonomics, Human information processing, Perception and attention, Memory and decision making, Mental workload	
Unit - V	**9 Hours**
IoT Applications in Ergonomics: wearable IoT devices for posture, motion, fatigue, heart rate, and EMG monitoring; environmental IoT sensors for workplace monitoring; real-time data acquisition and cloud-based monitoring; IoT-based ergonomic risk assessment and injury prevention; smart workstations and human–machine interaction; data analytics and visualization for ergonomics; ethical, privacy, and security issues in IoT-based ergonomics; industrial, office, and healthcare case studies.	

Ergonomics Laboratory

1. Experiments on fatigue measurement using bio-medical parameter.
2. Experiments on Measurement of anthropometric data.
3. Experiments on evaluation workstation.
4. Experiments on Measurement of local muscle activity using EMG
5. Experiments on virtual evaluation workstation.
6. Experiments on Human Computer Interactions.

Course Outcomes: After completing the course, the students will be able to:

CO1: Explain the fundamental concepts, principles, standards, and systems approach of ergonomics and their role in improving safety, comfort, efficiency, and human well-being in work systems.

CO2: Apply anthropometric data, posture analysis, and workstation design principles to design and evaluate seated and standing workplaces that accommodate human variability.

CO3: Analyse human–machine interaction, displays, controls, and environmental factors using ergonomic evaluation methods to assess and improve industrial, office, and service systems.
CO4: Demonstrate the use of IoT-based tools to assess ergonomic risks, monitor human performance, and propose data-driven ergonomic interventions.

References

[1] M. S. Sanders and E. J. McCormick, *Human Factors in Engineering and Design*, 7th ed. Singapore: McGraw-Hill, 1992, ISBN: 0071128263

[2] R. S. Bridger, *Introduction to Ergonomics*, 4th ed. Taylor & Francis, 2018, ISBN: 9781138582897

[3] G. Salvendy, Ed., *Handbook of Human Factors and Ergonomics*, 3rd ed. Hoboken, New Jersey, USA: John Wiley & Sons, 2006, ISBN: 9780471116904

[4] C. A. Phillips, *Human Factors Engineering*. New York: John Wiley & Sons, 2000, ISBN: 9780471370894

RUBRICS FOR THE CONTINUOUS INTERNAL EVALUATION		
#	**COMPONENTS**	**MARKS**
1	**QUIZZES:** Quizzes will be conducted in online/offline mode. TWO quizzes will be conducted and each Quiz will be evaluated for 10 Marks. **The sum of the two will be the final quiz marks.**	**20**
2	**TESTS:** Students will be evaluated in test through descriptive questions with different complexity levels (Revised Bloom's Taxonomy Levels: Remembering, Understanding, Applying, Analyzing, Evaluating, and Creating). THREE tests will be conducted (Two regular tests and One optional Improvement test). Each test will be evaluated for 50 Marks, adding upto 100 Marks. **Final test marks will be reduced to 40**.	**40**
3	**EXPERIENTIAL LEARNING:** Students shall be evaluated based on the creativity, innovation, and practical implementation of their proposed solutions. The evaluation shall be conducted in two phases. The implementation modalities for experiential learning activity shall be decided considering the nature of the course and its interdisciplinary requirements. **Each phase shall carry 20 marks, with the total evaluation carrying 40 marks.**	**40**
4	**LAB:** Conduction of laboratory exercises, lab report, observation and analysis (30 Marks), Lab test (10 Marks) and problem based learning review phase I and II (10 Marks) adding upto 50 marks . **The final lab marks is for 50.**	**50**
	MAXIMUM MARKS FOR THE CIE	**150**

RUBRICS FOR SEMESTER END EXAMINATION (THEORY)		
Q.NO.	**CONTENTS**	**MARKS**
	PART A	
1	Objective type questions covering entire syllabus	**10**
	PART B (Maximum of THREE Sub-divisions only)	
2	Unit - I: Compulsory	**18**
3	Unit - II: Compulsory	**18**

RUBRICS FOR SEMESTER END EXAMINATION (THEORY) (Continued)		
Q.NO.	**CONTENTS**	**MARKS**
4 & 5	Unit - III: Question 4 or 5	**18**
6 & 7	Unit - IV: Question 6 or 7	**18**
8 & 9	Unit - V: Question 8 or 9	**18**
	TOTAL	**100**

RUBRICS FOR SEMESTER END EXAMINATION (LAB)		
Q.NO.	**CONTENTS**	**MARKS**
1	Evaluation will be conducted as problem / project based learning, which is program / laboratory specific	**50**
	TOTAL	**50**

Semester - IV					
Operations Research **Category: Professional Core Course** **(Theory & Practical)**					
Course Code	**:**	IM244AI	**CIE**	**:**	100+ 50 Marks
Credit distribution L:T:P	**:**	3:0:1	**SEE**	**:**	100+ 50 Marks
Pedagogical Hours (L:T:P:S)	**:**	45:0:30:45	**CIE Duration**	**:**	2+2 Hours
Total credits:	**:**	4	**SEE Duration**	**:**	3+3 Hours

Unit - I	**9 Hours**
Introduction to Model Building: An Introduction to Modelling, Prescriptive or Optimization Models – Objective function, Decision Variable & Constraints, The Seven-Step Model-Building Process. **Introduction to Operations Research:** Definition of OR, Application of OR to Engineering and Managerial problems, Features of OR models, Limitations of OR. **Linear Programming:** Definition, Mathematical Formulation, Standard Form, Proportionality and Additivity Assumptions, Divisibility Assumption, Certainty Assumption, Feasible Region and Optimal Solution, Degenerate, A Diet Problem, A Work-Scheduling Problem, A Capital Budgeting Problem, Blending Problems, Production Process Models, The Graphical Solution of Two-Variable Linear Programming Problems.	
Unit - II	**9 Hours**
Simplex Algorithm: How to Convert an LP to Standard Form, Preview of the Simplex Algorithm, Direction of Unboundedness, Why Does an LP Have an Optimal basic feasible solution, The Simplex Algorithm, Using the Simplex Algorithm to Solve Minimization Problems, Alternative Optimal Solutions, Degeneracy and the Convergence of the Simplex Algorithm, The Big M Method, The Two-Phase Simplex Method. **Sensitivity Analysis and Duality:** A Graphical Introduction to Sensitivity Analysis, Some Important Formulas, Sensitivity Analysis, Finding the Dual of an Linear Programming, Economic Interpretation of the Dual Problem, The Dual Simplex Method.	
Unit - III	**9 Hours**
Transportation Problem: Formulating a transportation problem, General Description of a Transportation Problem, Basic feasible solution using different methods, Optimality Methods, Unbalanced transportation problem, Degeneracy in transportation problems, Variants in Transportation Problems, Applications of Transportation problems. **Assignment Problem:** Formulation of the Assignment problem, Solution method of assignment problem – Hungarian Method, Solution method of assignment problem – Hungarian Method, Variants in assignment problem, Travelling Salesman Problem. Usage of software tools to demonstrate Transportation and Assignment problems	
Unit - IV	**9 Hours**
Project Management Using Network Analysis: Network construction, CPM & PERT, Determination of critical path and duration, floats. Crashing of Network. Usage of software tools to demonstrate N/W flow problems	
Unit - V	**9 Hours**
Game Theory: Introduction, Two-person Zero Sum game, Pure strategies – Games with saddle point, Graphical Method, The rules of dominance, solution method of games without saddle point, Arithmetic method.	

Laboratory Work

1. Introduction to Operations Research Packages - using MAT Lab, GAMS Excel, TORA and LINGO
2. Exercise on application of Operations Research Models to various sector of economy including Manufacturing, Health Care, Infrastructure, Insurance, Banking, Retail, Agriculture and Governance

Course Outcomes: After completing the course, the students will be able to:

CO1: Formulate mathematical models for engineering and managerial problems.

CO2: Solve linear programming, transportation, and assignment problems using analytical and computational methods.

CO3: Construct and analyze project networks to optimize schedules using CPM, PERT, and software tools.

CO4: Analyze decision-making problems under uncertainty and determine optimal strategies using game theory.

References

[1] W. L. Winston, *Operations Research: Applications and Algorithms*, 4th ed. Thomson Books, 2025, ISBN: 9780357907818

[2] H. A. Taha, *Operations Research: An Introduction*, 11th ed. PHI Learning, 2025, ISBN: 9781292736099

[3] A. Ravindran et al., *Operations Research: Principles and Practice*, 2nd ed. John Wiley & Sons, 2007, ISBN: 9780471086086

[4] J. K. Sharma, *Operations Research: Theory and Applications*, 2nd ed. Pearson Education Pvt. Ltd., 2003, ISBN: 0333923944

RUBRICS FOR THE CONTINUOUS INTERNAL EVALUATION		
#	**COMPONENTS**	**MARKS**
1	**QUIZZES:** Quizzes will be conducted in online/offline mode. TWO quizzes will be conducted and each Quiz will be evaluated for 10 Marks. **The sum of the two will be the final quiz marks.**	**20**
2	**TESTS:** Students will be evaluated in test through descriptive questions with different complexity levels (Revised Bloom's Taxonomy Levels: Remembering, Understanding, Applying, Analyzing, Evaluating, and Creating). THREE tests will be conducted (Two regular tests and One optional Improvement test). Each test will be evaluated for 50 Marks, adding upto 100 Marks. **Final test marks will be reduced to 40**.	**40**
3	**EXPERIENTIAL LEARNING:** Students shall be evaluated based on the creativity, innovation, and practical implementation of their proposed solutions. The evaluation shall be conducted in two phases. The implementation modalities for experiential learning activity shall be decided considering the nature of the course and its interdisciplinary requirements. **Each phase shall carry 20 marks, with the total evaluation carrying 40 marks.**	**40**
4	**LAB:** Conduction of laboratory exercises, lab report, observation and analysis (30 Marks), Lab test (10 Marks) and problem based learning review phase I and II (10 Marks) adding upto 50 marks . **The final lab marks is for 50.**	**50**
	MAXIMUM MARKS FOR THE CIE	**150**

RUBRICS FOR SEMESTER END EXAMINATION (THEORY)		
Q.NO.	**CONTENTS**	**MARKS**
	PART A	
1	Objective type questions covering entire syllabus	**10**
	PART B (Maximum of THREE Sub-divisions only)	

RUBRICS FOR SEMESTER END EXAMINATION (THEORY) (Continued)		
Q.NO.	**CONTENTS**	**MARKS**
2	Unit - I: Compulsory	**18**
3	Unit - II: Compulsory	**18**
4 & 5	Unit - III: Question 4 or 5	**18**
6 & 7	Unit - IV: Question 6 or 7	**18**
8 & 9	Unit - V: Question 8 or 9	**18**
	TOTAL	**100**

RUBRICS FOR SEMESTER END EXAMINATION (LAB)		
Q.NO.	**CONTENTS**	**MARKS**
1	Evaluation will be conducted as problem / project based learning, which is program / laboratory specific	**50**
	TOTAL	**50**

Semester - IV					
Marketing Management **Category: Professional Core Course** **(Theory)**					
Course Code	**:**	IM245AT	**CIE**	**:**	100 Marks
Credit distribution L:T:P	**:**	3:0:0	**SEE**	**:**	100 Marks
Pedagogical Hours (L:T:P:S)	**:**	45:0:0:45	**CIE Duration**	**:**	2 Hours
Total credits:	**:**	3	**SEE Duration**	**:**	3 Hours

Unit - I	**7 Hours**
Introduction to Marketing: Marketing Management, Core Concepts of Marketing, Marketing Process Marketing mix - Marketing environment, Consumer Markets and buying behavior, Market segmentation and targeting and positioning, Marketing in the New Economy, Introduction to Digital Marketing	
Unit - II	**8 Hours**
Consumer Behavior: Concept and importance of consumer behavior; consumer buying decision process; factors influencing consumer behavior – cultural, social, personal, and psychological factors; buying roles and consumer involvement; types of buying behavior – complex, dissonance-reducing, habitual, and variety-seeking; post-purchase behavior and customer satisfaction;	
Unit - III	**7 Hours**
Content Marketing: Content Marketing Concepts & Strategies; Planning, Creating, Distributing & Promoting Content; Optimize Website UX & Landing Pages; Measure Impact; Metrics & Performance; Using Content Research for Opportunities, etc. **Search Engine Optimization:** Search Engine Optimization Fundamentals; Keywords and SEO Content Plan; SEO & Business Objectives; Writing SEO Content; On-site & off-site SEO; Optimize Organic Search Ranking, etc.	
Unit - IV	**7 Hours**
Social Media Marketing: Introduction; Major Social Media Platforms for Marketing; Developing Data-driven Audience & Campaign Insights; social media for Business; Creation & Optimization of Social Media Campaigns, etc. **Mobile Marketing:** Difference between mobile advertising and marketing, utilizing mobile marketing for sales promotions, online applications, etc. **E-mail Marketing:** Effective E-mail Campaigns; E-mail Plan; E-mail Marketing Campaign Analysis; Measuring Conversions & keeping up, etc.	
Unit - V	**7 Hours**
Conversion Optimization: What is AIDAS and its role; website optimization; what visitors want to see on the website; how to optimize key element and increase the effect of landing on a particular page **Web Analytics & Google Analytics:** Google Analytics Tools; Web Analytics Tools, etc. **Digital Analytics:** Evolution of Digital Analytics, information about end-to-end customer experience, analyst's influence on business, role as a change agent, etc.	
Course Outcomes: After completing the course, the students will be able to:	
CO1: Explain fundamental concepts of marketing management, consumer markets, and consumer buying behavior to understand marketing decision-making.	
CO2: Apply content marketing, SEO, social media, email, and mobile marketing tools to plan and execute digital marketing strategies.	
CO3: Analyze customer behavior, campaign performance, and conversion outcomes using web analytics, Google Analytics, and digital marketing metrics.	
CO4: Design and optimize digital assets such as websites, landing pages, and conversion funnels by integrating UX principles, the AIDAS model, and digital analytics to improve business performance.	
References	

[1] P. Kotler and K. L. Keller, *Marketing Management*, 15th ed. Pearson, 2016, ISBN: 9789332557185

[2] D. Chaffey and F. Ellis-Chadwick, *Digital Marketing: Strategy, Implementation and Practice*, 7th ed. Pearson, 2019, ISBN: 9781292241623

[3] D. S. Tull and D. I. Hawkins, *Marketing Research: Measurement and Method*, 6th ed. Prentice Hall India, 1993, ISBN: 8120309618

[4] P. Kotler et al., *Marketing Management: A South Asian Perspective*, 14th ed. Pearson, 2013, ISBN: 9788131767160

[5] D. A. Aaker et al., *Marketing Research*, 9th ed. John Wiley & Sons, 2008, ISBN: 9780470171912

RUBRICS FOR THE CONTINUOUS INTERNAL EVALUATION		
#	**COMPONENTS**	**MARKS**
1	**QUIZZES:** Quizzes will be conducted in online/offline mode. TWO quizzes will be conducted and each Quiz will be evaluated for 10 Marks. **The sum of the two will be the final quiz marks.**	**20**
2	**TESTS:** Students will be evaluated in test through descriptive questions with different complexity levels (Revised Bloom's Taxonomy Levels: Remembering, Understanding, Applying, Analyzing, Evaluating, and Creating). THREE tests will be conducted (Two regular tests and One optional Improvement test). Each test will be evaluated for 50 Marks, adding upto 100 Marks. **Final test marks will be reduced to 40**.	**40**
3	**EXPERIENTIAL LEARNING:** Students shall be evaluated based on the creativity, innovation, and practical implementation of their proposed solutions. The evaluation shall be conducted in two phases. The implementation modalities for experiential learning activity shall be decided considering the nature of the course and its interdisciplinary requirements. **Each phase shall carry 20 marks, with the total evaluation carrying 40 marks.**	**40**
	MAXIMUM MARKS FOR THE CIE	**100**

RUBRICS FOR SEMESTER END EXAMINATION (THEORY)		
Q.NO.	**CONTENTS**	**MARKS**
	PART A	
1	Objective type questions covering entire syllabus	**10**
	PART B (Maximum of THREE Sub-divisions only)	
2	Unit - I: Compulsory	**18**
3	Unit - II: Compulsory	**18**
4 & 5	Unit - III: Question 4 or 5	**18**
6 & 7	Unit - IV: Question 6 or 7	**18**
8 & 9	Unit - V: Question 8 or 9	**18**
	TOTAL	**100**

Semester - IV					
Semiconductor Manufacturing and Process Technology **Category: Professional Core Elective** **(Theory)**					
Course Code	**:**	ME246TAA	**CIE**	**:**	100 Marks
Credit distribution L:T:P	**:**	3:0:0	**SEE**	**:**	100 Marks
Pedagogical Hours (L:T:P:S)	**:**	45:0:0:45	**CIE Duration**	**:**	2 Hours
Total credits:	**:**	3	**SEE Duration**	**:**	3 Hours

Unit - I	**9 Hours**
Integrated Circuits: First transistor, integrated circuit, Moore's law, Feature and wafer size, Definition of the integrated circuit technology node. Overview of Integrated Circuits, Manufacturing materials, Processing equipment, Metrology tools, Circuit design, Mask formation, Wafer processing, Basic Structure of an Integrated Circuit Fabrication facility.	
Unit - II	**9 Hours**
Wafer Fabrication: Introduction, Crystal Structures and Defects, Crystal orientation, Sand to Wafer, Crude silicon, Silicon purification, Crystal pulling, Czochralski method, Floating zone method, Wafering, Wafer finishing	
Unit - III	**9 Hours**
Lithography: Introduction, Photoresist, Photolithography process, Wafer cleaning, preparation, Photoresist coating, soft bake, Alignment and exposure, post exposure bake, Development, Hard bake, Metrology and defect inspection	
Unit - IV	**9 Hours**
Semiconductor Packaging and testing: Introduction to Semiconductor Packaging, importance in Modern Electronics. Packaging Technologies: leaded and leadless packages, surface mount technology, and ball grid array, General flow of IC testing, Wafer test and IC package test, functional test, Build-in-self-test, Burn-in test	
Unit - V	**9 Hours**
Cleanroom Technology: air conditioning and refrigeration equipment; air locks and air showers, floor systems, disinfection and sterilization; Performance Requirements for Clean Room Garments, Filtration Mechanisms, Filter types	

Course Outcomes: After completing the course, the students will be able to:

CO1: Understand the step-by-step sequence of processes used in creating various layers of a semiconductor chip.

CO2: Identify the environment requirements essential for high-quality semiconductor manufacturing

CO3: Evaluate fabricated devices to assess desirable properties.

CO4: Apply principles of integrated circuit (IC) design and layout to develop simple electronic devices

References

[1] H. Xiao, *Introduction to Semiconductor Manufacturing Technology*, 2nd ed. SPIE Press, 2018, ISBN: 9781510616530

[2] G. S. May and C. J. Spanos, *Fundamentals of Semiconductor Manufacturing and Process Control*. John Wiley & Sons, 2000, ISBN: 9780471784067

[3] D. K. Schroder, *Semiconductor Material and Device Characterization*, 3rd ed. John Wiley & Sons, 2006, ISBN: 9780471739067

[4] J. D. Plummer, *Integrated Circuit Fabrication*. Cambridge University Press, 2023, ISBN: 9781009303583

RUBRICS FOR THE CONTINUOUS INTERNAL EVALUATION		
#	**COMPONENTS**	**MARKS**
1	**QUIZZES:** Quizzes will be conducted in online/offline mode. TWO quizzes will be conducted and each Quiz will be evaluated for 10 Marks. **The sum of the two will be the final quiz marks.**	**20**
2	**TESTS:** Students will be evaluated in test through descriptive questions with different complexity levels (Revised Bloom's Taxonomy Levels: Remembering, Understanding, Applying, Analyzing, Evaluating, and Creating). THREE tests will be conducted (Two regular tests and One optional Improvement test). Each test will be evaluated for 50 Marks, adding upto 100 Marks. **Final test marks will be reduced to 40**.	**40**
3	**EXPERIENTIAL LEARNING:** Students shall be evaluated based on the creativity, innovation, and practical implementation of their proposed solutions. The evaluation shall be conducted in two phases. The implementation modalities for experiential learning activity shall be decided considering the nature of the course and its interdisciplinary requirements. **Each phase shall carry 20 marks, with the total evaluation carrying 40 marks.**	**40**
	MAXIMUM MARKS FOR THE CIE	**100**

RUBRICS FOR SEMESTER END EXAMINATION (THEORY)		
Q.NO.	**CONTENTS**	**MARKS**
	PART A	
1	Objective type questions covering entire syllabus	**10**
	PART B (Maximum of THREE Sub-divisions only)	
2	Unit - I: Compulsory	**18**
3	Unit - II: Compulsory	**18**
4 & 5	Unit - III: Question 4 or 5	**18**
6 & 7	Unit - IV: Question 6 or 7	**18**
8 & 9	Unit - V: Question 8 or 9	**18**
	TOTAL	**100**

Semester - IV					
Facilities Planning and Material Handling Systems **Category: Professional Core Elective** **(Theory)**					
Course Code	**:**	IM246TAB	**CIE**	**:**	100 Marks
Credit distribution L:T:P	**:**	3:0:0	**SEE**	**:**	100 Marks
Pedagogical Hours (L:T:P:S)	**:**	45:0:0:45	**CIE Duration**	**:**	2 Hours
Total credits:	**:**	3	**SEE Duration**	**:**	3 Hours

Unit - I	**9 Hours**
Introduction to Facilities Planning & Digital Transformation: Facilities planning: objectives, scope, Facility location and layout types, Productivity and flow considerations, Introduction to Industry 4.0 and smart factories, Role of IoT, Cyber-Physical Systems (CPS), and data-driven decision making, Digital transformation in manufacturing and service facilities	
Unit - II	**9 Hours**
Systematic Layout Planning (SLP) with Digital Tools: Muther's SLP methodology, Activity relationship charts and flow analysis, From-to charts and space requirements, Development of block layouts, Computer-aided layout design tools, Introduction to CAD-based facility planning.	
Unit - III	**9 Hours**
Quantitative Techniques & Simulation: Load-distance model, Assignment model and QAP, Heuristic methods (CRAFT, CORELAP), Simulation modeling for facility layout (overview), Optimization using metaheuristics (GA, PSO – basics), Data-driven layout evaluation.	
Unit - IV	**9 Hours**
Smart Material Handling Systems: Principles of material handling, Equipment selection and unit load concept, Automated systems: AGVs, AMRs, conveyors, AS/RS, IoT-enabled tracking and RFID systems, Warehouse automation and smart logistics, Cost analysis and performance evaluation.	
Unit - V	**9 Hours**
Digital Facility Design & Future Trends: Ergonomics and safety in digital environments, Sustainable and green facility design, Digital Twin in facility planning, AI/ML applications in layout optimization, Cloud-based facility management systems, Human-centric design (Industry 5.0), Basics of VR/AR for facility visualization.	

Course Outcomes: After completing the course, the students will be able to:

CO1: Explain facility planning concepts, layout types, and digital transformation in manufacturing systems

CO2: Apply SLP and quantitative techniques using digital tools for facility design

CO3: Analyze and select material handling systems including automated and smart systems

CO4: Design and evaluate facilities using digital technologies considering sustainability, ergonomics, and cost

References

[1] J. A. Tompkins et al., *Facilities Planning*, 4th ed. Hoboken, NJ, USA: John Wiley & Sons, 2010, ISBN: 9780470444047

[2] S. S. Heragu, *Facilities Design*, 4th ed. Boca Raton, FL, USA: CRC Press, 2016, ISBN: 9781498732901

[3] F. E. Meyers and M. P. Stephens, *Manufacturing Facilities Design and Material Handling*, 4th ed. Upper Saddle River, NJ, USA: Pearson, 2013, ISBN: 9780133027792

[4] S. V. Nath and P. van Schalkwyk, *Building Industrial Digital Twins: Design, Develop, and Deploy Digital Twin Solutions for Real-World Industries*. Birmingham, U.K.: Packt Publishing, 2021, ISBN: 9781803233993

[5] A. Hemmatian and Y. Najibifar, *Digital Twins in Construction: From Design to Facility Management*. Canada: Heritage Branch Publishing, 2025

[6] B. Atkin and A. Brooks, *Total Facility Management*, 5th ed. Hoboken, NJ, USA: Wiley-Blackwell, 2021, ISBN: 9781119703952

[7] International Facility Management Association (IFMA), *Facility management trends and best practices*, Houston, TX, USA, 2020

[8] J. Lee et al., "A cyber-physical systems architecture for industry 4.0-based manufacturing systems," *Manufacturing Letters*, vol. 3, pp. 18–23, 2015. DOI: 10.1016/j.mfglet.2014.12.001

[9] O. Hakimi et al., "Digital twin-enabled smart facility management: A review," *Engineering Management*, 2024

RUBRICS FOR THE CONTINUOUS INTERNAL EVALUATION		
#	**COMPONENTS**	**MARKS**
1	**QUIZZES:** Quizzes will be conducted in online/offline mode. TWO quizzes will be conducted and each Quiz will be evaluated for 10 Marks. **The sum of the two will be the final quiz marks.**	**20**
2	**TESTS:** Students will be evaluated in test through descriptive questions with different complexity levels (Revised Bloom's Taxonomy Levels: Remembering, Understanding, Applying, Analyzing, Evaluating, and Creating). THREE tests will be conducted (Two regular tests and One optional Improvement test). Each test will be evaluated for 50 Marks, adding upto 100 Marks. **Final test marks will be reduced to 40**.	**40**
3	**EXPERIENTIAL LEARNING:** Students shall be evaluated based on the creativity, innovation, and practical implementation of their proposed solutions. The evaluation shall be conducted in two phases. The implementation modalities for experiential learning activity shall be decided considering the nature of the course and its interdisciplinary requirements. **Each phase shall carry 20 marks, with the total evaluation carrying 40 marks.**	**40**
	MAXIMUM MARKS FOR THE CIE	**100**

RUBRICS FOR SEMESTER END EXAMINATION (THEORY)		
Q.NO.	**CONTENTS**	**MARKS**
	PART A	
1	Objective type questions covering entire syllabus	**10**
	PART B (Maximum of THREE Sub-divisions only)	
2	Unit - I: Compulsory	**18**
3	Unit - II: Compulsory	**18**
4 & 5	Unit - III: Question 4 or 5	**18**
6 & 7	Unit - IV: Question 6 or 7	**18**

RUBRICS FOR SEMESTER END EXAMINATION (THEORY) (Continued)		
Q.NO.	**CONTENTS**	**MARKS**
8 & 9	Unit - V: Question 8 or 9	**18**
	TOTAL	**100**

Semester - IV					
Fundamentals of Sustainability Engineering and Circular Economy **Category: Professional Core Elective** **(Theory)**					
Course Code	**:**	IM246TAC	**CIE**	**:**	100 Marks
Credit distribution L:T:P	**:**	3:0:0	**SEE**	**:**	100 Marks
Pedagogical Hours (L:T:P:S)	**:**	45:0:0:45	**CIE Duration**	**:**	2 Hours
Total credits:	**:**	3	**SEE Duration**	**:**	3 Hours

Unit - I	**9 Hours**
Basics of Sustainable Engineering: Introduction, Why and What is Sustainable Development, THE SDGs, Paris Agreement and Role of Engineering, Sustainable Development and the Engineering Profession, Key attributes of the Graduate Engineering **Concepts and Principles of Sustainable Engineering:** Goals of sustainability, System Thinking, Life Cycle Thinking and Circular Economy, Green Economy and Low Carbon Economy, Eco Efficiency, Triple bottom Line, Guiding principles of sustainable engineering, Frameworks for sustainable Engineering.	
Unit - II	**9 Hours**
Fundamentals of Life Cycle Assessment: Why and What is LCA, LCA Goal and Scope, Life cycle inventory, Life Cycle Impact Assessment, Interpretation and presentation of Results, Iterative Nature of LCA, Methodological Choices, LCI Databases and LCA Software's, Strength and Limitations of LCA. Social Impact & Life Cycle Costing, Life Cycle Cost Analysis tools **Life Cycle Assessment for Decision Making:** How LCA helps in Decision Making, Future Prospects and Sustainable Assessment.	
Unit - III	**9 Hours**
Engineering Sustainability, Needs for Metrology and Standards: Introduction, Providers of Standards and Methods of Development, Standards, A Conceptual Standard Framework, Corporate **Sustainability Sustainability and Future Trends:** Introduction, Sustainable Operations and Supply Chain Management, The Impact of Climate Change, Demographic Evolution, Urbanization, Emerging New Consumers, Smart Technologies and the Digital Factory, Mega-Trends,	
Unit - IV	**9 Hours**
Introduction to Circular Economy: Towards Circular Economy, What's the Need. Principles of Circular Economy, Understanding the Concepts, Framework, Design, UN SDGs & Circular Economy **Circular Economy and Engineering:** Integrating Sustainable Engineering, Circular Economy in Engineering Discipline, Circular Eco -Design (3Rs: Re -think, Re -Design & Re -Purpose), Circular Eco Design using the 7R model	
Unit - V	**9 Hours**
Circular Economy and Environment: Impact on Environment Policies, Aids in Resource Management, Impact on Economic and Social Indicators, Advantages of Circular economy over Linear Economy, Circular Economy and Sustainability, Sectoral and Business Case studies of Circular Economy	
Course Outcomes: After completing the course, the students will be able to:	
CO1: Examine the foundational concepts of sustainable development, while evaluating the ethical role of the engineering profession in achieving a Green and Low-Carbon Economy.	
CO2: Apply the methodological framework of Life Cycle Assessment to evaluate the environmental, social, and economic costs of engineering systems	
CO3: Appraise the impact of mega-trends on corporate sustainability, metrology standards, and global supply chain operations.	
CO4: Synthesize circular economy frameworks to transition from linear to circular systems, utilizing case studies to design resource-efficient engineering solutions	

References

[1] T. Ramjeawon, *Introduction to Sustainability for Engineers*, 1st ed. CRC Press, 2020, ISBN: 9780367254452

[2] P. Modak, *Practicing Circular Economy*. CRC Press, 2020, ISBN: 9780367619572

[3] V. Belvedere and A. Grando, *Sustainable Operations and Supply Chain Management*. John Wiley & Sons, 2017, ISBN: 9781119285366

[4] S. K. Sikdar et al., *Measuring Progress Towards Sustainability: A Treatise for Engineers*. Springer International Publishing, 2016, ISBN: 9783319427195

RUBRICS FOR THE CONTINUOUS INTERNAL EVALUATION		
#	**COMPONENTS**	**MARKS**
1	**QUIZZES:** Quizzes will be conducted in online/offline mode. TWO quizzes will be conducted and each Quiz will be evaluated for 10 Marks. **The sum of the two will be the final quiz marks.**	**20**
2	**TESTS:** Students will be evaluated in test through descriptive questions with different complexity levels (Revised Bloom's Taxonomy Levels: Remembering, Understanding, Applying, Analyzing, Evaluating, and Creating). THREE tests will be conducted (Two regular tests and One optional Improvement test). Each test will be evaluated for 50 Marks, adding upto 100 Marks. **Final test marks will be reduced to 40**.	**40**
3	**EXPERIENTIAL LEARNING:** Students shall be evaluated based on the creativity, innovation, and practical implementation of their proposed solutions. The evaluation shall be conducted in two phases. The implementation modalities for experiential learning activity shall be decided considering the nature of the course and its interdisciplinary requirements. **Each phase shall carry 20 marks, with the total evaluation carrying 40 marks.**	**40**
	MAXIMUM MARKS FOR THE CIE	**100**

RUBRICS FOR SEMESTER END EXAMINATION (THEORY)		
Q.NO.	**CONTENTS**	**MARKS**
	PART A	
1	Objective type questions covering entire syllabus	**10**
	PART B (Maximum of THREE Sub-divisions only)	
2	Unit - I: Compulsory	**18**
3	Unit - II: Compulsory	**18**
4 & 5	Unit - III: Question 4 or 5	**18**
6 & 7	Unit - IV: Question 6 or 7	**18**
8 & 9	Unit - V: Question 8 or 9	**18**
	TOTAL	**100**

Semester - IV					
Database Management Systems **Category: Professional Core Elective** **(Theory)**					
Course Code	**:**	IM246TAD	**CIE**	**:**	100 Marks
Credit distribution L:T:P	**:**	3:0:0	**SEE**	**:**	100 Marks
Pedagogical Hours (L:T:P:S)	**:**	45:0:0:45	**CIE Duration**	**:**	2 Hours
Total credits:	**:**	3	**SEE Duration**	**:**	3 Hours

Unit - I	**9 Hours**
Databases and Database Users: Introduction, characteristics of data base approach, intended uses of a DBMS, advantages and implication of database approach. **Database Systems Concepts and Architecture:** Data models, schemas and instances, DBMS architecture and data independence, database languages and interfaces, database system environment, classification of data base management systems.	
Unit - II	**9 Hours**
Data Modeling: High level conceptual data models for database design. Entity types, entity sets, attributes, and keys. Relationships, relationship types, roles, and structural constraints. Weak entity types. ER diagrams and design issues.	
Unit - III	**9 Hours**
Relational Data Model and Relational Algebra: Brief discussion on CODD rules, relational model concepts, constraints, and schemas. Update operation on relations, basic and additional relational algebra operations, and queries in relational algebra.	
Unit - IV	**9 Hours**
Structured Query Language (SQL): Data definition etc. in SQL2. Basic and complex queries in SQL. Insert, delete, update statements, and views in SQL, embedded SQL.	
Unit - V	**9 Hours**
Database Design: Design guidelines for relational schemas, functional Dependencies, normalization -1st, 2nd, 3rd, 4th and 5th normal forms. **NOSQL Databases and Big Data Storage Systems:** Introduction to NOSQL Systems, The CAP Theorem, Document-Based NOSQL Systems and MongoDB, NOSQL Key-Value Stores, NOSQL Graph Databases and Neo4j.	

Course Outcomes: After completing the course, the students will be able to:

CO1: Describe the terminologies and associated concepts embodied in database systems.

CO2: Illustrate appropriate data models for the design of database system.

CO3: Demonstrate an understanding of the design of relational databases.

CO4: Formulate solutions to a broad range of query and data update problems.

References

[1] R. Elmasri and S. B. Navathe, *Fundamentals of Database Systems*, 7th ed. Pearson Education, 2016, ISBN: 9780133970777

[2] R. Ramakrishnan and J. Gehrke, *Database Management Systems*, 3rd ed. McGraw Hill Education, 2014, ISBN: 9789339213114

[3] A. Kahate, *Introduction to Database Management Systems*, 1st ed. Pearson India, 2004, ISBN: 9788131700785

[4] G. W. Hansen and J. V. Hansen, *Database Management and Design*. Prentice-Hall of India Pvt. Ltd., 2005, ISBN: 9788120314658

RUBRICS FOR THE CONTINUOUS INTERNAL EVALUATION		
#	COMPONENTS	MARKS
1	**QUIZZES:** Quizzes will be conducted in online/offline mode. TWO quizzes will be conducted and each Quiz will be evaluated for 10 Marks. **The sum of the two will be the final quiz marks.**	**20**
2	**TESTS:** Students will be evaluated in test through descriptive questions with different complexity levels (Revised Bloom's Taxonomy Levels: Remembering, Understanding, Applying, Analyzing, Evaluating, and Creating). THREE tests will be conducted (Two regular tests and One optional Improvement test). Each test will be evaluated for 50 Marks, adding upto 100 Marks. **Final test marks will be reduced to 40**.	**40**
3	**EXPERIENTIAL LEARNING:** Students shall be evaluated based on the creativity, innovation, and practical implementation of their proposed solutions. The evaluation shall be conducted in two phases. The implementation modalities for experiential learning activity shall be decided considering the nature of the course and its interdisciplinary requirements. **Each phase shall carry 20 marks, with the total evaluation carrying 40 marks.**	**40**
	MAXIMUM MARKS FOR THE CIE	**100**

RUBRICS FOR SEMESTER END EXAMINATION (THEORY)		
Q.NO.	CONTENTS	MARKS
	PART A	
1	Objective type questions covering entire syllabus	**10**
	PART B (Maximum of THREE Sub-divisions only)	
2	Unit - I: Compulsory	**18**
3	Unit - II: Compulsory	**18**
4 & 5	Unit - III: Question 4 or 5	**18**
6 & 7	Unit - IV: Question 6 or 7	**18**
8 & 9	Unit - V: Question 8 or 9	**18**
	TOTAL	**100**

Semester - IV					
Applied Design Thinking Lab **Category: Professional Core Course** **(Practical)**					
Course Code	**:**	IM248DL	**CIE**	**:**	50 Marks
Credit distribution L:T:P	**:**	0:0:1	**SEE**	**:**	50 Marks
Pedagogical Hours (L:T:P:S)	**:**	0:0:30:0	**CIE Duration**	**:**	1.5 Hours
Total credits:	**:**	1	**SEE Duration**	**:**	2 Hours

Guidelines for Design Thinking Lab:

1. The Design Thinking Lab (DTL) is to be carried out by a team of two–three students.
2. Each student in a team must contribute equally to the tasks mentioned below.
3. Each group has to select a theme that provides solutions to challenges of societal concern. Normally, three to four themes will be identified by the department.
4. Each group should follow the stages of Empathy, Design, Ideate, Prototype, and Test for successful completion of the DTL.
5. After every stage of the DTL, the committee constituted by the department, along with the coordinators, will conduct the Continuous Internal Evaluation (CIE). The committee shall consist of the respective coordinator and two senior faculty members as examiners. The evaluation will be carried out separately for each student.
6. The team should prepare a digital poster, and a final report shall be submitted after incorporating the modifications, if any, suggested by the evaluation committee.

The Design Thinking lab tasks would involve: 1. Carry out the detailed questionnaire

7. Carry out a detailed questionnaire to identify the problem under the selected theme. Prepare an empathy report based on the responses obtained from the stakeholders.
8. For the identified problem, the team shall propose an innovative, out-of-the-box solution to successfully complete the Ideation stage of the Design Thinking Lab (DTL).
9. Once the solution idea is finalized, formulate a detailed design during the Design stage, considering its practical feasibility and implementation aspects.
10. After approval of the design, implement the proposed solution and develop a working prototype of the system.
11. Conduct thorough testing of all modules of the developed prototype and perform integrated testing to validate its overall functionality.
12. Demonstrate the functioning of the prototype along with a presentation explaining the design, implementation, and testing outcomes.
13. Prepare a digital poster illustrating each stage of the Design Thinking Lab (DTL). Submit a detailed project report highlighting the activities carried out, along with the difficulties and challenges encountered at every stage of the DTL.

Course Outcomes: After completing the course, the students will be able to:

CO1: Apply design thinking principles to identify and define societal problems through stakeholder engagement and need analysis.

CO2: Analyze user requirements and generate innovative solutions using creative problem-solving techniques.

CO3: Evaluate design alternatives through prototyping, testing, validation, and performance assessment.

CO4: Develop sustainable and user-centric prototypes and communicate design outcomes through reports, presentations, and digital posters.

RUBRICS FOR THE CONTINUOUS INTERNAL EVALUATION		
#	**COMPONENTS**	**MARKS**
1	**Presentation 1** Selection of topic- (phase 1) Justification for Importance, need of the hour with surveyed data	**50**
	MAXIMUM MARKS FOR THE CIE	**50**

RUBRICS FOR SEMESTER END EXAMINATION (LAB)		
Q.NO.	**CONTENTS**	**MARKS**
1	Written presentation of synopsis: Write up (05), Presentation of the project (15), Demonstration of the project (20), Viva (05), Report (05), **Total marks added upto 50**.	**50**
	TOTAL	**50**

Semester - IV					
Bridge Course C Programming **Category: Bridge Course** **(Common to All Programs)** **(Theory)**					
Course Code	**:**	CS149BT	**CIE**	**:**	50 Marks
Credit distribution L:T:P	**:**	2:0:0	**SEE**	**:**	—
Pedagogical Hours (L:T:P:S)	**:**	30:0:0:30	**CIE Duration**	**:**	1.5 Hours
Total credits:	**:**	2	**SEE Duration**	**:**	—

Unit - I	**10 Hours**

Introduction to Programming: Introduction to Computer Software, Classification of Computer Software, Programming Languages, Generation of Programming Languages,Problem-solving using Algorithms and Flowcharts. **Introduction to C:** Introduction,structure of a C program, Writing the first program, Files used in a C program. Compiling and executing C Programs using comments, C Tokens, Character set in C, Keywords,Identifiers, Basic Data Types in C, Variables, Constants, I/O statements in C. Types of errors in C.

Unit - II	**10 Hours**

Operators in C: Arithmetic, Relational, Logical, Assignment, Increment and Decrement, Conditional operator, Bitwise, sizeof, and Comma operator. Type conversion and type casting. **Decision Control and Looping Statement:** Introduction to decision control statements, conditional branching statements, iterative statements, Nested loops, Break and continue statements, goto statements. **Arrays:** Introduction, Declaration of Arrays, Accessing elements of an array, Storing values in arrays, Operations on 1D Arrays-Traversal, Insertion and Deletion of element in an array. Two dimensional arrays-Operations on two dimensional arrays.

Unit - III	**10 Hours**

Strings: Introduction, Operations on strings- finding length of a string, converting characters of a string into uppercase and lowercase, concatenating two strings, appending a string to another string, comparing two string, reversing a string. String and character Built in functions. **Functions:** Introduction, Using functions, Function declaration/function prototype, Function definition, Function call, Return statement, Passing parameters to a function, Built-in functions. Passing arrays to functions, Scope of variables, Storage Classes, Recursive Functions. **Structures:** Introduction: Structure Declaration, Typedef declaration, initialization of structures, accessing members of a structures. **Pointers:** Introduction to pointers, declaring pointer variables.

Course Outcomes: After completing the course, the students will be able to:

CO1: Understand the fundamental concepts of C programming, including program structure, data types, operators, and control statements, to develop simple C programs

CO2: Apply decision-making and looping constructs to solve computational problems using structured and logically correct C programs

CO3: Design and implement modular programs using functions, including recursion, by applying appropriate variable scope and parameter passing techniques.

CO4: Analyze and implement arrays, strings, pointers, and structures to solve problem-oriented programming tasks in C.

References

[1] R. Thareja, *Programming in C*. Oxford University Press, 2023

[2] E. Balagurusamy, *Programming in ANSI C*. McGraw Hill Education (India) Pvt. Ltd., 2019

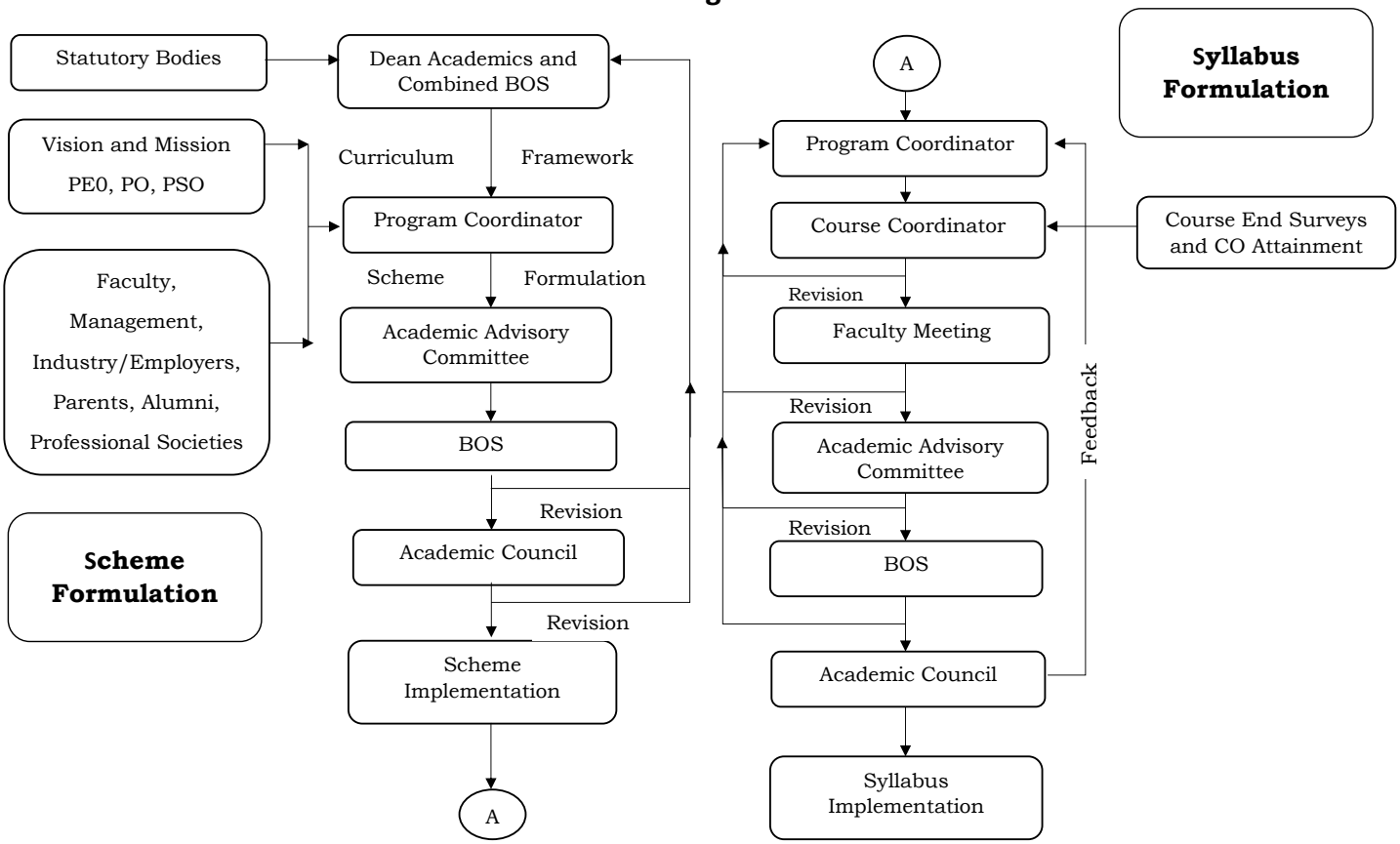
[3] H. Schildt, *C: The Complete Reference*, 4th ed. Berkeley, Calif.: Osborne/McGraw-Hill, 2000, 805 pp., Previous ed.: 1995, ISBN: 0072121246

[4] R. C. Backhouse, *Algorithmic Problem Solving*. Chichester, UK: Wiley, 2011, 414 pp., Includes bibliographical references and index

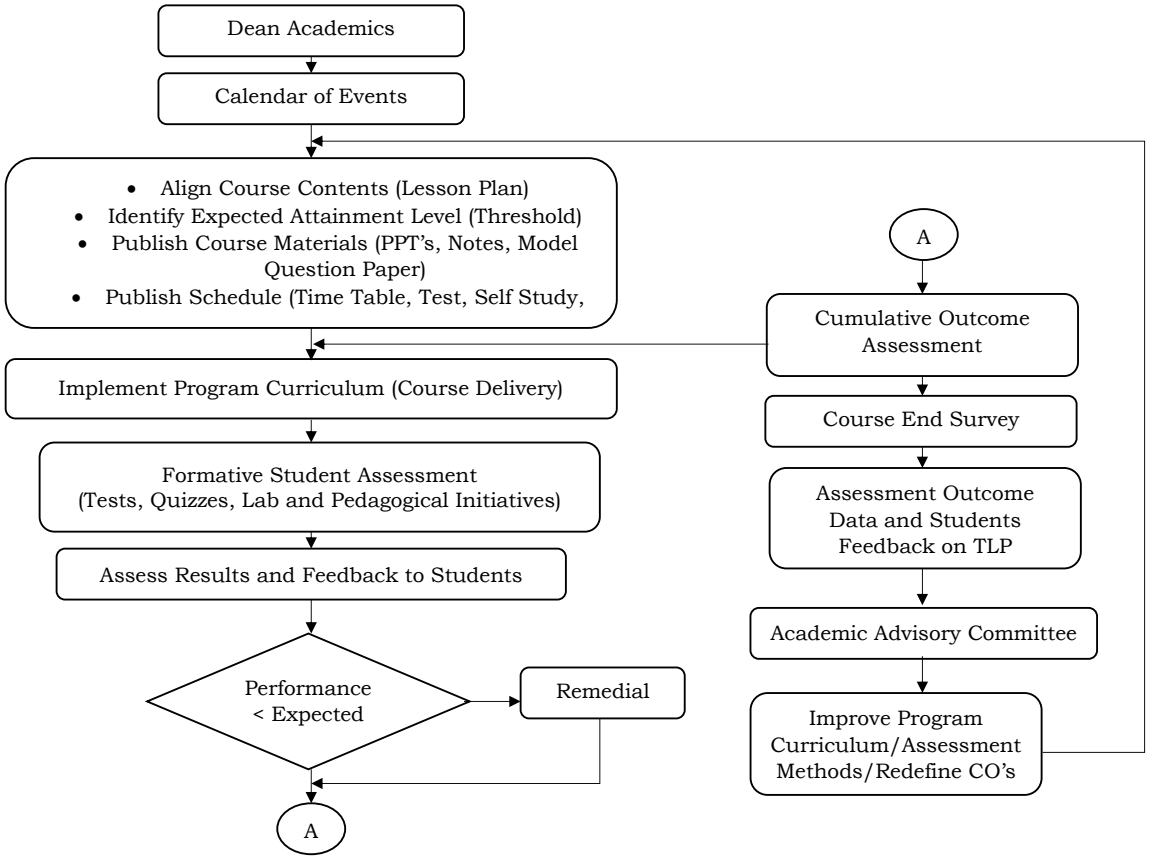
[5] B. W. Kernighan and D. M. Ritchie, *The C Programming Language* (Prentice-Hall Software Series), 2nd ed. Upper Saddle River, NJ: Prentice-Hall PTR, 2014, 272 pp., ISBN: 0131103709

RUBRICS FOR THE CONTINUOUS INTERNAL EVALUATION		
#	**COMPONENTS**	**MARKS**
1	**QUIZZES:** Quizzes will be conducted in online/offline mode. **TWO QUIZZES** will be conducted and each Quiz will be evaluated for 10 Marks. **THE SUM OF TWO WILL BE THE FINAL QUIZ MARKS.**	**10**
2	**TESTS:** Students will be evaluated in test, descriptive questions with different complexity levels (Revised Bloom's Taxonomy Levels: Remembering, Understanding, Applying, Analyzing, Evaluating, and Creating). TWO tests will be conducted. Each test will be evaluated for 50 Marks, adding upto 100 Marks. **FINAL TEST MARKS WILL BE REDUCED TO 40 MARKS.**	**20**
3	**EXPERIENTIAL LEARNING:** Students will be evaluated for their creativity and practical implementation of the problem. Case stufy-based teaching learning (05), Program specific requirements (05), Video based seminar/presentation/demonstration (10), Tools usage (20). **ADDING UPTO 40 MARKS.**	**20**
	MAXIMUM MARKS FOR THE CIE	**50**

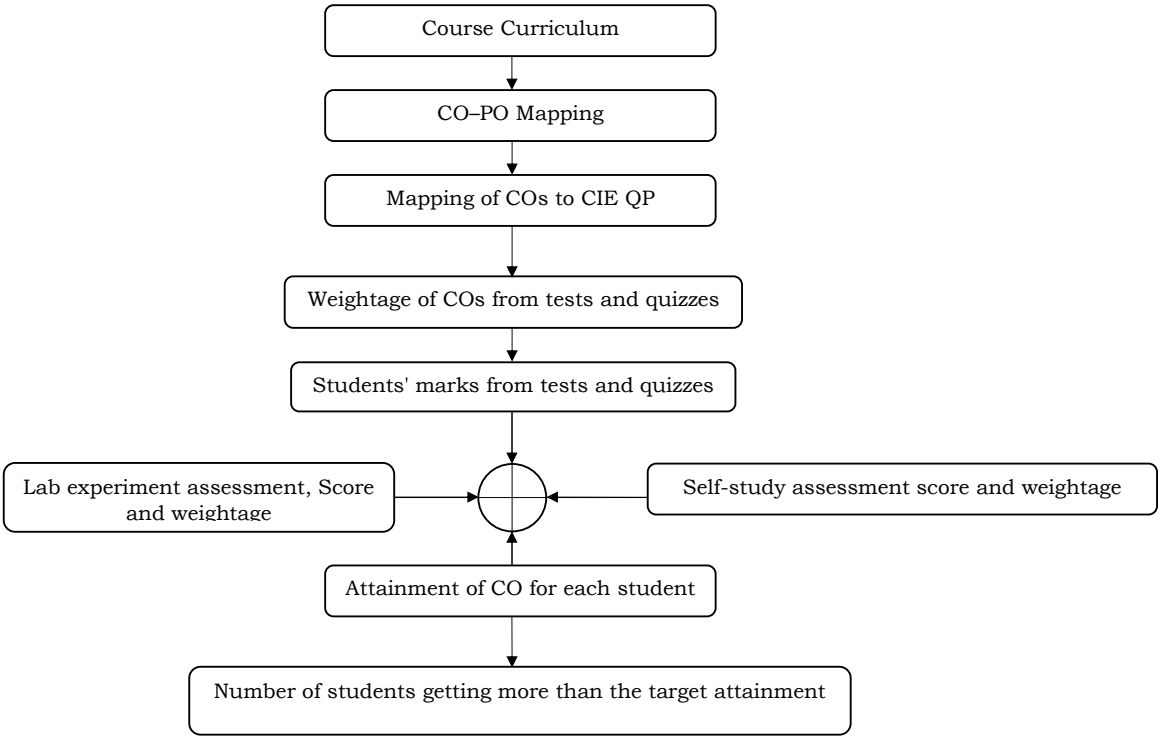
Curriculum Design Process



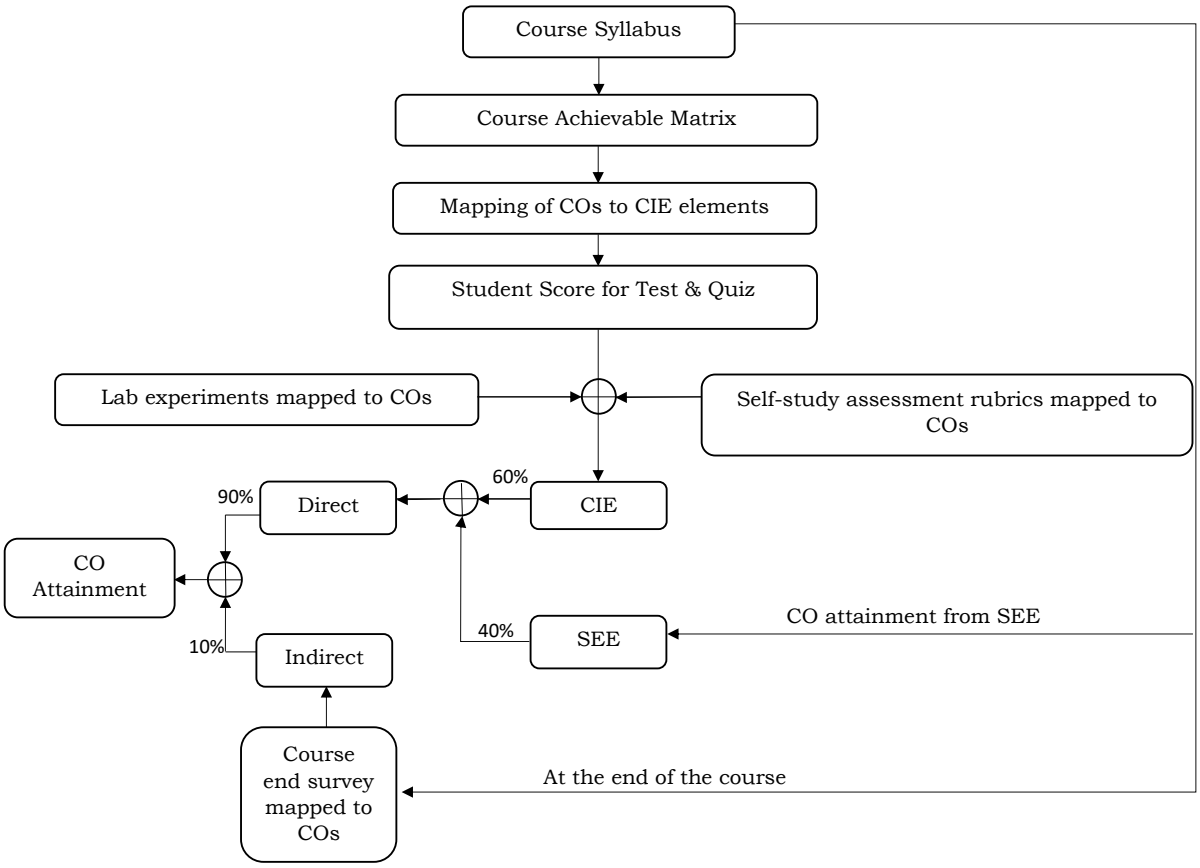
Academic Planning and Implementation



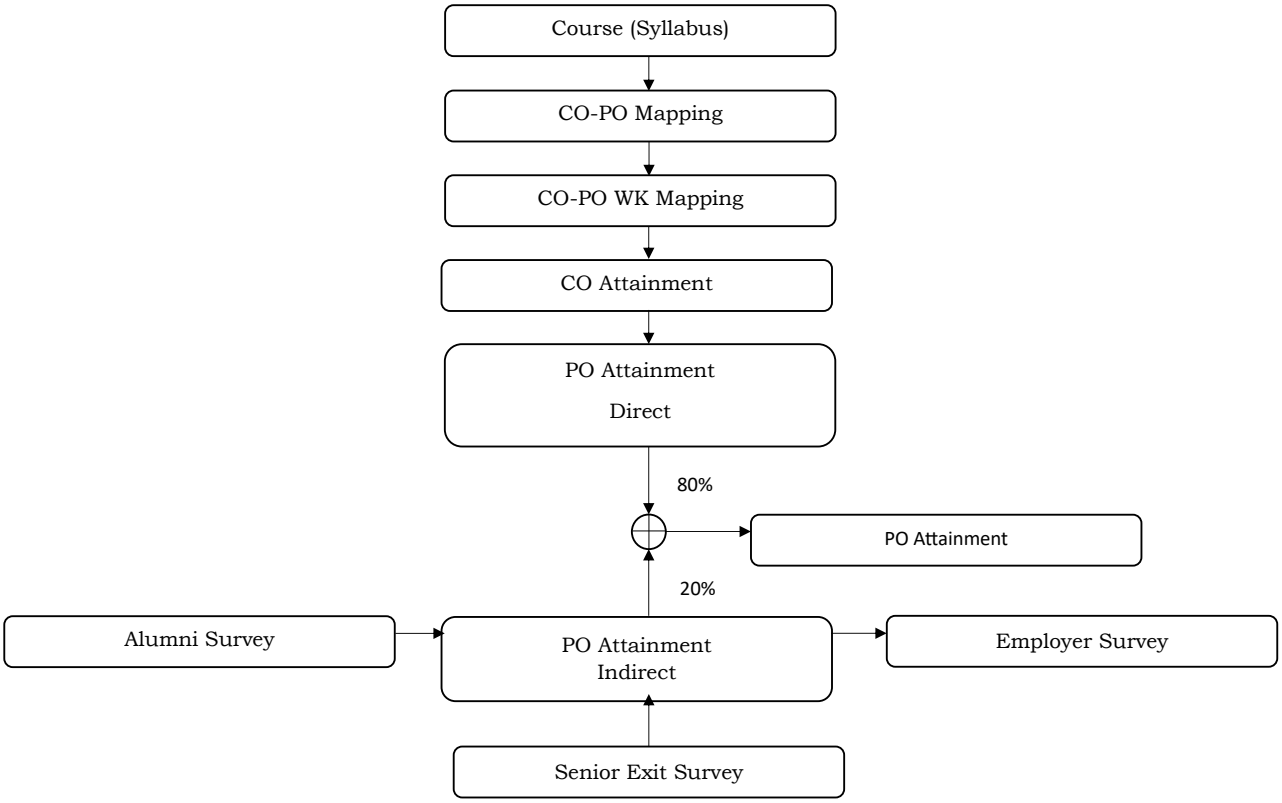
Process for Course Outcome Attainment



Final CO Attainment Process



Program Outcome Attainment Process



KNOWLEDGE & ATTITUDE PROFILE

WK1: A systematic, theory-based understanding of the natural sciences applicable to the discipline and awareness of relevant social sciences.

WK2: Conceptually-based mathematics, numerical analysis, data analysis, statistics and formal aspects of computer and information science to support detailed analysis and modelling applicable to the discipline.

WK3: A systematic, theory-based formulation of engineering fundamentals required in the engineering discipline.

WK4: Engineering specialist knowledge that provides theoretical frameworks and bodies of knowledge for the accepted practice areas in the engineering discipline; much is at the forefront of the discipline.

WK5: Knowledge, including efficient resource use, environmental impacts, whole-life cost, re-use of resources, net zero carbon, and similar concepts, that supports engineering design and operations in a practice area.

WK6: Knowledge of engineering practice (technology) in the practice areas in the engineering discipline.

WK7: Knowledge of the role of engineering in society and identified issues in engineering practice in the discipline, such as the professional responsibility of an engineer to public safety and sustainable development.

WK8: Engagement with selected knowledge in the current research literature of the discipline, awareness of the power of critical thinking and creative approaches to evaluate emerging issues.

WK9: Ethics, inclusive behaviour and conduct. Knowledge of professional ethics, responsibilities, and norms of engineering practice. Awareness of the need for diversity by reason of ethnicity, gender, age, physical ability etc. with mutual understanding and respect, and of inclusive attitudes.

PROGRAM OUTCOMES (POs)

PO1: Engineering Knowledge: Apply knowledge of mathematics, natural science, computing, engineering fundamentals and an engineering specialization as specified in WK1 to WK4 respectively to develop the solution of complex engineering problems.

PO2: Problem Analysis: Identify, formulate, review research literature and analyze complex engineering problems reaching substantiated conclusions with consideration for sustainable development. (WK1 to WK4)

PO3: Design/Development of Solutions: Design creative solutions for complex engineering problems and design/develop systems/components/processes to meet identified needs with consideration for the public health and safety, whole-life cost, net zero carbon, culture, society and environment as required. (WK5)

PO4: Conduct Investigations of Complex Problems: Conduct investigations of complex engineering problems using research-based knowledge including design of experiments, modelling, analysis & interpretation of data to provide valid conclusions. (WK8).

PO5: Engineering Tool Usage: Create, select and apply appropriate techniques, resources and modern engineering & IT tools, including prediction and modelling recognizing their limitations to solve complex engineering problems. (WK2 and WK6)

PO6: The Engineer and The World: Analyze and evaluate societal and environmental aspects while solving complex engineering problems for its impact on sustainability with reference to economy, health, safety, legal framework, culture and environment. (WK1, WK5, and WK7).

PO7: Ethics: Apply ethical principles and commit to professional ethics, human values, diversity and inclusion; adhere to national & international laws. (WK9)

PO8: Individual and Collaborative Team work: Function effectively as an individual, and as a member or leader in diverse/multi-disciplinary teams.

PO9: Communication: Communicate effectively and inclusively within the community and society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations considering cultural, language, and learning differences.

PO10: Project Management and Finance: Apply knowledge and understanding of engineering management principles and economic decision-making and apply these to one's own work, as a member and leader in a team, and to manage projects in multidisciplinary environments.

PO11: Life-long Learning: Recognize the need for, and have the preparation and ability to i) independent and life-long learning ii) adaptability to new and emerging technologies and iii) critical thinking in the broadest context of technological change. (WK8)

INNOVATIVE TEAMS OF RVCE

Ashwa Mobility Foundation (AMF): Designs and fabricates Formula-themed race cars and mobility solutions to address urban transportation issues.

Astra Robotics Team: Focuses on designing and building application-specific robots.

Coding Club: Helps students gain coding skills and succeed in competitions like GSoC and ACM-ICPC.

Frequency Club Team: Works on software and hardware, emphasizing AI and Machine Learning.

Team Garuda: Develops a supermileage urban concept electric car and E-mobility products.

Team Jatayu: Builds low-cost UAVs with autonomous capabilities for various tasks.

Team Antariksh: Focuses on space technology and the development of operational rockets.

Team Chimera: Builds a Formula Electric Car through R&D in E-Mobility.

Helios Racing Team: Designs and tests All-Terrain Vehicles, participating in SAE's BAJA competitions.

Team Krushi: Creates low-cost farming equipment to assist farmers in cultivation and harvesting.

Team Vyoma: Designs and tests radio-controlled aircraft and UAVs.

Team Dhruva: Engages in astronomy-related activities and collaborates on projects with organizations like ICTS and IIA.

Ham Club: Promotes Amateur Radio and explores technical innovations in communications, especially for disaster response.

Chitrak : It is the official electric bike team of RV College of Engineering (RVCE) in Bangalore

Anoraniya:A Quantum based technical club initiated by students. The club is dedicated to addressing various technical challenges in quantum technologies and is on the verge of delivering some fascinating results.

Accelerate Club: organizes and participates in hackathons, bootcamps, workshops, and student-led innovation drives.

Team Hydra: Develops autonomous underwater vehicles for applications like water purification and waste detection.

Entrepreneurship Development Cell (EDC): Builds entrepreneurial skills through courses, mentorship, and startup-focused innovation initiatives.

DHI Club: Promotes learning and project development in Artificial Intelligence and Machine Learning.

SPARK (IUCEE Student Chapter): Encourages project-based learning and interdisciplinary collaboration beyond classroom academics.

Google Developer Groups (GDG): Supports student developers through technical learning, innovation activities, and global community engagement.

Women in Cloud: Empowers women in technology through mentorship, cloud computing education, and networking opportunities.

ACM RVCE Student Chapter: Connects students with the global computing community through technical events and research exposure.

Cultural Activity Teams

- ALAAP
- Studio Zero
- CARV English
- CARV Hindi
- CARV Kannada
- Sattva
- F6.3 Photography club
- RAAG
- Rotaract Club of RVCE
- DebSoc
- QuizCorp RV
- Footprints
- Evoke
- TEDXRVCE

NSS of RVCE

NCC of RVCE



VISION

Leadership in Quality Technical Education, Interdisciplinary Research & Innovation, with a Focus on Sustainable and Inclusive Technology



MISSION

- To deliver outcome based Quality education, emphasizing on experientiallearning with the state of the art infrastructure.
- To create a conducive environment for interdisciplinary research and innovation.
- To develop professionals through holistic education focusing on individual growth, discipline, integrity, ethics and social sensitivity.
- To nurture industry-institution collaboration leading to competency enhancement and entrepreneurship.
- To focus on technologies that are sustainable and inclusive, benefiting all sections of the society.



QUALITY POLICY

Achieving Excellence in Technical Education, Research and Consulting through an Outcome Based Curriculum focusing on Continuous Improvement and Innovation by Benchmarking against the global Best Practices.



CORE VALUES

Professionalism, Commitment, Integrity, Team Work, Innovation